



A Survey of Multi-modal Knowledge Graphs: Technologies and Trends

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In recent years, Knowledge Graphs (KGs) have played a crucial role in the development of advanced knowledge-intensive applications, such as recommender systems and semantic search. However, the human sensory system is inherently multi-modal, as objects around us are often represented by a combination of multiple signals, such as visual and textual. Consequently, Multi-modal Knowledge Graphs (MMKGs), which combine structured knowledge representation with multiple modalities, represent a powerful extension of KGs. Although MMKGs can handle certain types of tasks (e.g., visual query answering) or queries that standard KGs cannot process, and they can effectively tackle some standard problems (e.g., entity alignment), we lack a widely accepted definition of MMKG. In this survey, we provide a rigorous definition of MMKGs along with a classification scheme based on how existing approaches address four fundamental challenges: representation, fusion, alignment, and translation, which are crucial to improving an MMKG. Our classification scheme is flexible and allows for easy incorporation of new approaches, as well as a comparison of two approaches in terms of how they address one of the fundamental challenges mentioned above. As the first comprehensive survey of MMKG, this article aims at inspiring and provide a reference for relevant researchers in the field of Artificial Intelligence.

CCS Concepts: • **Hardware** → *Neural systems*;

Additional Key Words and Phrases: Multi-modal knowledge graphs, four fundamental challenges, pre-training in MMKGs

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1 INTRODUCTION

Knowledge Graphs (KGs), or simply KGs, are a critical component of many knowledge-based applications, including semantic search [105], recommender systems [113], and question answering [32]. Although KGs were first introduced in 1972 [81], they gained widespread popularity in 2012 when Google¹ announced its KG as a tool to significantly enhance the capabilities of its search engine [84]. In layman's terms, a KG consists of a collection of triplets of the form $\langle e_1, r, e_2 \rangle$, where e_1 and e_2 identify real-world objects, also known as *entities* and r specifies the type of *relationship* that binds e_1 and e_2 . Several KGs are available for commercial and academic use, including Freebase [6], DBpedia [2], and YAGO [88]. Most of the existing KGs are built from textual sources, and advanced algorithms are available to extract entities and their relationships from the text. However, an entity may be associated with signals of a different nature, such as one or more images accompanying a textual description. Therefore, relying solely on textual information would not allow us to benefit from complementary signals describing the entity itself.

In what follows, we refer to each of the signals associated with an entity as a *modality*, and we begin our discussion by noting that the human cognitive system is inherently *multi-modal*: for example, basic everyday tasks such as object recognition often require the acquisition and processing of different signals (e.g., we see an object and we touch it). On the one hand, the use of multiple modalities brings our knowledge representation process closer to the human cognitive model. On the other hand, the availability of multi-modal data offers unprecedented opportunities for advanced reasoning tasks and for the design of innovative applications that would not be possible if only a single modality were available.

To clarify this point, let us consider a key problem in KGs known as **Entity Alignment (EA)** [22, 90, 112]. In short, EA involves determining whether two entities in the same KG (or possibly in two different KGs) refer to the same real-world object. A large number of EA algorithms use a structural approach [59, 90]. This involves defining and constructing the neighborhood of an entity, and then computing the degree of matching between neighborhoods to make a decision. However, structural approaches have limitations, particularly if text is the only source available for encoding knowledge. For example, suppose we construct a KG from a collection of texts in different languages. Documents written in one language may provide rich descriptions of an entity and its neighborhood, while documents in another language may provide limited information about the same entity. Therefore, it would be extremely difficult to match entities referring to the same real-world object but encoded in different languages. The availability of images to supplement text could be a decisive tool for solving the entity alignment task. Images are universal and easy to understand, making them a useful semantic bridge for matching entities that would otherwise be regarded as distinct objects. Furthermore, software agents could use multiple modalities associated with the same entity to answer more sophisticated queries. For example, given a picture of a mobile phone, a software agent could integrate heterogeneous signals to guess the brand of the phone. Similarly, given a picture of a painting, the agent could leverage different modalities to guess the artist who authored the painting.

The combination of structured knowledge provided by standard KGs, along with the availability of multi-modal data, has given rise to a novel tool for representing knowledge known as **Multi-modal Knowledge Graphs (MMKGs)**. In Figure 1, we present a fragment of an MMKG, where entities such as the city of San Francisco, the State of California, and the Golden Gate Bridge are represented as colored circles, and relationships are depicted through arrows. We also note that some entities, such as the State of California, are accompanied by audio and video information. The

¹<https://www.google.com/>

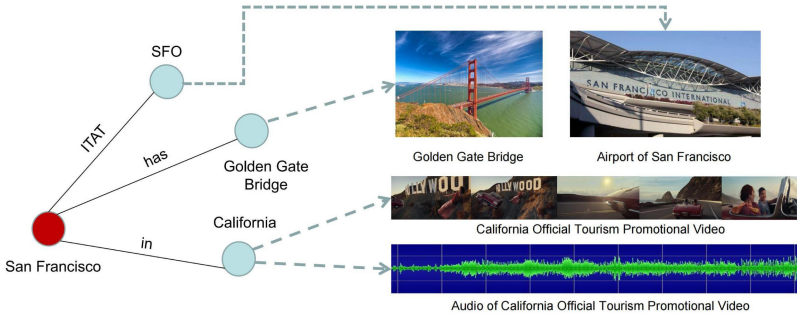


Fig. 1. A small fragment of a MMKG. Entities are represented as colored circles and arrows specifies relationships among available entities. Each entity could be associated with one or more pieces of information, possibly in different modalities.

concept of MMKGs is relatively new and largely unexplored. A quick search on Google Trends² using keywords such as “Multi-modal,” “Knowledge Graph,” “Multi-modal Learning,” and “Multi-modal Knowledge Graph” shows that the trend associated with the MMKG topic is almost zero in the period ranging from 2021 to 2022, but it starts increasing from 2022 onwards. The first survey devoted to constructing MMKGs is reported in [119], and the number of articles (and their citations) about MMKGs is small. Furthermore, the name “MMKG” is not universally known and is often confused with terms such as “multimedia database” or “multi-modal database”.

In this article, we aim at addressing the aforementioned shortcomings by providing three main reasons for doing so. Firstly, research on MMKGs is not well-known to a large audience of researchers and engineers. We hope that our survey will contribute toward increasing awareness and popularity of MMKGs. Secondly, while significant research has been done in the field of machine learning algorithms applied to the analysis of multi-modal data (as demonstrated in the excellent article by Baltrusaitis et al. [4]), it is important to note that MMKGs are distinct from multi-modal machine learning. While multi-modal machine learning algorithms are relevant for generating rich entity representations that are useful for constructing MMKGs, the true strength of MMKGs lies in their ability to represent knowledge in a structured manner and perform advanced reasoning tasks on this structured knowledge. We expect that our survey will help researchers from different fields understand the definition of MMKGs, the methods used to construct them from raw data, and the models used to complete them. Thirdly, we believe there is much room for improvement in the performance of MMKGs. Our survey aims at identifying current research problems as well as promising research avenues that deserve consideration in the future.

To follow up on the motivations above, we provide a rigorous definition of what MMKGs are (and are not). We believe that a sound definition can help researchers and practitioners understand when MMKGs are useful and when they are not. Precisely defining MMKGs also enables us to differentiate them from neighboring disciplines, such as multi-modal machine learning. This differentiation allows us to assess whether achievements in related research areas contribute to advancing the state of the art in MMKGs, and vice versa, and how open problems in the MMKG research area can foster research in other disciplines. We have also classified and summarized the available methods for constructing MMKGs. Our contribution can help scientists and engineers fairly evaluate the pros and cons of each method and select the most appropriate method to construct an MMKG in a given application domain.

²<https://trends.google.com/trends/?geo=US>

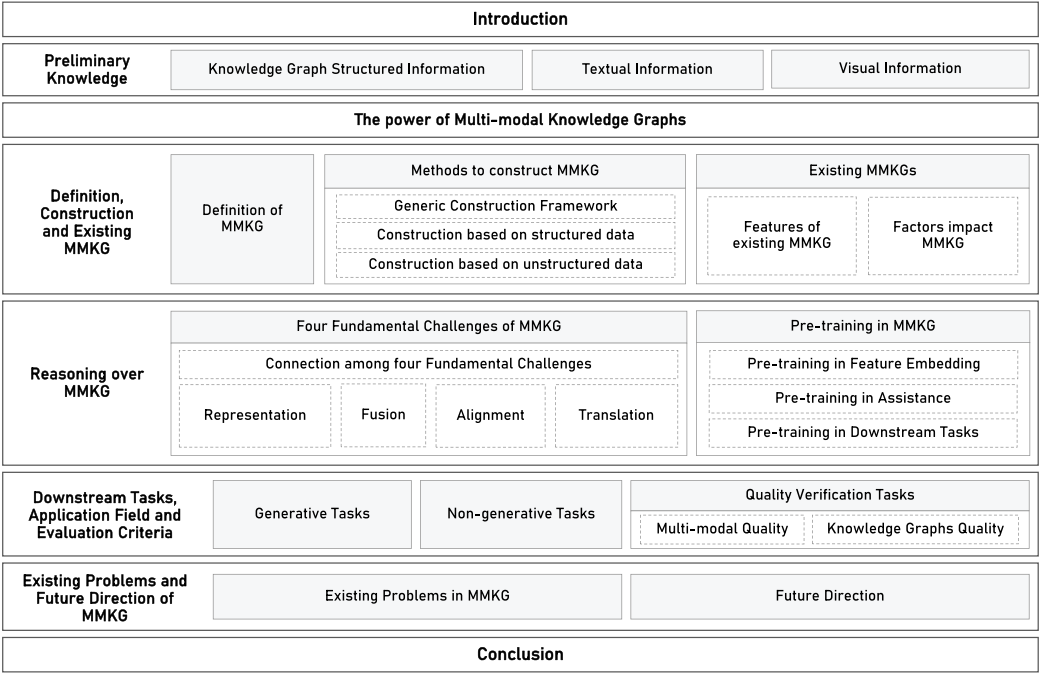


Fig. 2. Structure of this survey.

Table 1. Acronym Table

Acronym	Meaning	Acronym	Meaning
MMKG	Multi-modal Knowledge Graph	NER	Named Entity Recognition
BERT	Bidirectional Encoder Representation from Transformers	GPT	Generative Pre-Training
CNN	Convolution Neural Network	MLP	Multilayer Perceptron
RNN	Recurrent Neural Network	GNN	Graph Neural Network
RBM	Restricted Boltzmann Machines	GCN	Graph Convolution Network
CBOW	Continuous Bag of Words	HMM	Hidden Markov Models
VGG	Visual Geometry Group (Model)	MRFs	Markov Random Fields
YOLO	You Only Look Once (Model)	CRF	Conditional Random Field
RCNN	Region Convolution Nerual Network	KB	Knowledge Base
ELMo	Embeddings from Language Models	SVM	Support Vector Machines
CRF	Conditional Random Fields	NLP	Natural Language Processing
GloVe	Global Vectors for Word Representation	ResNet	Residual neural Network
FCN	Fully Convolutional Networks	SIF	Sommth Inverse Frequency
OCR	Optical Character Recognition	SGD	Stochastic Gradient Descent
GAN	Generative Adversarial Networks	GATs	Graph Attention neTwork
AMA	Associative Multi-channel Auto-Encoders	PoE	Products of Experts
BLEU	BiLingual Evaluation Understudy	MI	Mutual Information
METEOR	Metric for Evaluation of Translation with Explicit ORdering	RBF	Radial Basis Function
BPR	Bayesian Personalized Ranking	SVM	Support Vector Machine
FLAT	Chinese NER Using Flat-Lattice Transformer	BCE	Binary Cross Entropy
BART	Bidirectional and Auto-Regressive Transformers	CW	Contextualized Word
LHGC	Louvain Hierarchical Graph Clustering	RMSE	Root Mean Square Error

At present, the number of approaches to constructing and reasoning about MMKGs is relatively small, and the number of available MMKG datasets is limited. Therefore, it is difficult to establish a taxonomy of existing approaches and generalize any possible taxonomy to include future MMKG approaches. To this end, we identified four major challenges in the field of MMKG: representation, fusion, alignment, translation. These challenges are also core technical challenges in the field of

multi-modal machine learning, but there are significant differences in how these challenges are treated in the context of MMKGs. We reviewed existing MMKG approaches by pointing out how existing methods address one or more of these challenges. Our approach to classifying the existing literature has two advantages: we gain a deep understanding of the principles underlying proposed MMKG models, and our taxonomy is flexible enough to easily accommodate new approaches. Additionally, we discuss the use of pre-trained models in MMKGs due to their recent success in many Natural Language Processing applications. We focus on open problems and limitations in MMKGs, and we identify improvement methods as well as future development directions for MMKGs. We believe that the identification of existing problems and promising research avenues can inspire researchers and engineers.

The main contributions of this article are three folds:

- As far as we know, this article is the first comprehensive investigation into the MMKG and its technical details.
- Our exposition of MMKG is comprehensive and systematic, covering its definition, construction methods, four major fundamental challenges, pre-training, downstream tasks, and quality evaluation methods in detail. We also provide a detailed analysis of each dataset and model, enabling relevant researchers to understand their specifics.
- We have identified issues with MMKG and conducted an in-depth discussion of recommended improvements and future development directions.

The structure of this survey is graphically shown in Figure 2. In Section 2, we introduce preliminary background material on MMKG (including knowledge graphs, textual information, and visual information) which will be extensively used in our article. Section 3 explains what the potential of MMKGs is and why they enable us to perform tasks that would not otherwise be possible on traditional KGs. In Section 4, we give a formal definition of MMKG along with a description of the construction methods of MMKG. In Section 5, we delve into the reasoning over MMKG, including the four fundamental challenges we cited before, that is representation, fusion, alignment, and translation. We also emphasize the impact of pre-training techniques on MMKG. In Section 6, we introduce the downstream tasks, application fields, and evaluation criteria of MMKG. In Section 7, we address the existing problems and future directions of MMKG. Finally, we draw our conclusions in Section 8. Given this article's frequent use of acronyms, we provide an acronym table, i.e., Table 1.

2 PRELIMINARY KNOWLEDGE

In this section, we introduce some background material that will be extensively used throughout the papers, covering the unimodal embedding techniques for knowledge graph structured information, textual information, and visual information.

2.1 Knowledge Graph Structured Information

The structured representation of available information lays the foundation for constructing an MMKG and reasoning over it. To represent the graph structure of the knowledge graph, we first need to use a scoring function to measure the rationality of the triple facts, and then use an encoding model to map entities and relationships to the selected representation space. In this section, we will elaborate on the process of representing structured information in knowledge graphs based on *representation space*, *scoring functions*, and *encoding model*.

Representation Space. Representation Space refers to the process of mapping entities and relationships in a knowledge graph onto vectors of a multidimensional space, called *embeddings*. Embedding vectors can be used to calculate the similarity or distance between entities, making

them a key component for enabling KG reasoning tasks. Point-wise space, complex vector space, Gaussian space, and manifold and group space are common representation spaces, with point-wise space being the most widely used. In knowledge graph modeling, the choice of a suitable representation space is a key step, as different representation spaces can be used to handle entities and relationships of varying complexities. Modeling choices may have significant impacts on the performance and effectiveness of the knowledge graph. The TransE approach [7] and its variant model, TransX series, are the most popular methods for constructing a representation space, which we will discuss in detail in the following.

Scoring Functions. Scoring functions include two typical types: *distance-based* functions and *similarity-based* functions.

Distance-based scoring functions measure the credibility of the facts by calculating the distance between head and tail entities. Bordes et al. [7] proposed the TransE model, a translation-based model for knowledge graph embedding. For each fact triple (h, r, t) , TransE represents entities and relations in the same space and it regards the relation vector r as the translation between the head entity vector h and the tail entity vector t , namely, it constructs r to satisfy the following expression $h + r \approx t$. TransE uses margin-based ranking loss to optimize the embedding parameters, which encourages the correct triple to have a higher score than any corrupted triple. TransE is a simple and effective method but the translation assumption we described above can negatively affect its accuracy when it deals with complex relationships such as *one-to-many* or *many-to-many* relationships. In order to solve the limitations of TransE in complex relationships, TransH [97] introduces a hyperplane to model a relation, allowing an entity to have different representations under different relationships. In addition, TransH introduces advanced methods to sample negative triples. A further interesting approach is TransR, due to Lin et al. [47] which uses separate spaces to model entities and relationships and it assumes that an entity consists of multiple attributes. The idea of TransR is that different relationships focus on different attributes of the entity and different relations have different semantic spaces. Lin et al. proposed an extension of TransR called CTransR: for a specific relation r , all entity pairs (h, t) in the training data are then clustered into multiple groups, under the constraint that entity pairs in each group are expected to be tied with r . TransD [33], proposed by Ji et al., takes into account both the variety of relations and the variety of entities, distinguishing it from the TransR/CTransR model. Similar to TransD, TransG [100] clusters pre-trained relations to solve the problem of multiple semantic expressions of relations. TransSparse [34] simplifies TransR by enforcing sparsity on the projection matrix. TransA [99], proposed by Xiao et al., incorporates a symmetric non-negative matrix M_r for each relation r and utilizes the adaptive Mahalanobis distance as the scoring function. By learning the distance metric M_r , TransA is more flexible when dealing with complex relations. PTransE [46] mainly models multi-hop relational paths, which can be used for knowledge reasoning.

Scoring methods based on semantic similarity assess the reliability of facts by evaluating their semantic alignment, often employing a multiplicative matrix formulation. Prominent examples of techniques belonging to this category include DistMult [108] and HoIE [107].

Encoding Models. Encoding models are essential in knowledge representation and reasoning because they help encode entity relation data into meaningful vector spaces. Over the years, researchers have developed different types of encoding models, including linear/bilinear models [73], factorization models [78], and neural networks [111].

Linear/bilinear models were some of the earliest encoding models developed and they have the goal to capture interactions between entities and relations by representing them in linear or bilinear form.

Factorization models [78] are encoding models that use matrix factorization techniques to decompose entity-relation matrices into low-dimensional embeddings. These models can

handle sparse and incomplete data and they can capture complex relationships between entities.

Neural networks have become increasingly popular in recent years because of their ability to capture complex non-linear relationships between entities and their strong expressive power. Different types of neural networks commonly used in encoding models include **Multilayer Perceptrons (MLP)** [31], **Convolutional Neural Networks (CNNs)** [8], **Recurrent Neural Networks (RNNs)** [111], **Transformers** [92], **Graph Neural Networks (GNNs)** [79], and **Restricted Boltzmann Machines (RBMs)** [64]. The usage of CNNs for vector mapping ensures good performances and, in particular, convolutional models like ConvE [17], ConvKB [65], and ConvR [36] have been successfully proposed to capture complex relationships between entities and relations. In detail, ConvE first converts head entities and relations into 2D vectors and then uses convolutional and fully connected layers to obtain interaction information. As for ConvKB, it represents each triple as a 3-column matrix and feeds it into a convolutional layer with multiple filters to generate different feature maps. The R-GCN model [80] extends graph convolutional networks to use polyadic encoding for entity embeddings. ConvR improves upon ConvE by using 2D convolution to represent entities: such a choice is effective to reduce the number of parameters as well as to enrich interactions between entities and relations.

2.2 Textual Information

Text information representation refers to the vectorization of textual information. The representation of text information enables machines to understand textual descriptions, which is fundamental for some downstream tasks (such as text generation and question answering). According to [95], text representation methods can be classified into static and dynamic. In the following, we describe in detail both the two representation methods.

Static Representation. refers to a set of parameters or features used in a system or model to represent invariant or relatively stable characteristics. These parameters (or features) are typically determined during the training process of the system and used as inputs during prediction. Static representation mainly uses either sparse high-dimensional vectors or dense low-dimensional vectors to represent text. Sparse high-dimensional vectors are represented by one-hot encoding while low-dimensional vector representations include SENNA [14], CBOW, Skip-gram, GloVe [69], and so on. Word2Vec [61], a word embedding method proposed by Google in 2013, learns semantic knowledge with an unsupervised method from a large text corpus. Word2Vec includes CBOW, which predicts the current value from context, and skip-gram, which uses the current word to predict context. **Global Vectors for Word Representation (GloVe)** [69] is a word embedding technique that leverages both count-based and global statistical information. It assigns a vector to each word using real-valued numbers, capturing various semantic relationships between words, such as similarity, analogy, and more.

Dynamic Representation. refers to a set of parameters or features used to represent the changing state of a system or model. Compared to static representation, dynamic representation is more flexible because it can adapt to changes in the system or model's state. Dynamic representation is commonly used for processing sequence data such as time series, natural language text, audio signals, and more. Dynamic representation often utilizes structures such as RNNs or **Long Short-Term Memory networks (LSTMs)** to model context information in a sequence. Dynamic representation has been an essential aspect of recent developments in natural language processing. Relevant approaches to producing dynamic representations include CoVe [60], ELMo [70], GPT [72], and BERT [18]. ELMo uses a bidirectional LSTM model to pre-train the language model while fine-tuning the output based on the input text. ELMo has been shown to be effective in various natural language processing tasks such as sentiment analysis, natural language inference,

and named entity recognition. BERT stands for Bidirectional Encoder Representations from Transformers. The idea behind BERT is similar to ELMo, but it uses the Transformer network for pre-training. BERT has emerged as a highly successful model, surpassing previous benchmarks in numerous natural language processing tasks. Its exceptional performance spans diverse domains such as question-answering, sentiment analysis, and language inference. GPT is another prominent dynamic representation model developed by OpenAI. Unlike ELMo and BERT, GPT uses a generative model trained on a large corpus of text data. GPT has demonstrated impressive performance in language generation tasks such as language translation and text completion.

2.3 Visual Information

Visual embedding is a crucial technique in computer vision, involving the mapping of pixel information from images or videos to low-dimensional vector representations. This mapping is typically achieved through the use of deep CNNs, which excel at preserving visual features of images [8]. Over the years, researchers have proposed a number of CNN-based models for various visual recognition tasks. For example, the VGG network was introduced in 2014 and it stacked multiple layers of 3×3 small convolution kernels and 2×2 maximum pooling layers to achieve impressive results (see [83]). In the same year, Girshick et al. proposed the R-CNN method [23], which divided images into regions based on texture similarity and performed object detection on them. Experimental trials prove that the R-CNN architecture was expensive to train and it had slow training speeds. In 2015, the **Residual Neural Network (ResNet)** was introduced: ResNet stacked CNNs in multiple layers and added shortcut links to overcome the degradation problem [30]. In 2016, the Faster R-CNN model was introduced, which was based on pre-trained models and fine-tuned for training [76]. Mask R-CNN was also introduced in the same year, which added **Fully Convolutional Networks (FCNs)** [53] to Faster R-CNN to generate the corresponding mask branch [29]. In 2016, the simple and fast YOLO model was proposed [75]. Unlike Faster R-CNN, YOLO can use the global image in detection to learn the general information of the target, which has a certain universality [75]. In 2014, Goodfellow et al. proposed the GAN, which contain a *generative* and a *discriminative* model [24]. The generative model generates natural-looking instances similar to the original data, while the discriminative model judges whether a given instance appears to be naturally real or artificial.

The architecture based on CNN has long been dominant in image processing, but in recent years, some researchers have started applying Transformers, which have been successful in natural language processing, to images. One such method is known as **Visual Transformer (ViT)** [20]: the image is divided into fixed-size blocks, akin to words in a sentence. Each block undergoes linear embedding and positional embedding is added to retain positional information, before being input into the Transformer for training. Experiments have shown that after training on large datasets, ViT can be successfully transferred to smaller tasks and achieve excellent results.

Advancements in visual embedding have greatly improved the performance of computer vision systems, allowing them to tackle a wide range of visual tasks with unprecedented accuracy and efficiency. The availability of effective tools for generating visual embeddings, as will be seen in the following sections, is crucial for the accurate construction of an MMKG.

3 THE POWER OF MULTI-MODAL KNOWLEDGE GRAPHS

One natural question that arises is about the power of Multi-Modal Knowledge Graphs against standard Knowledge Graphs. In this section we briefly illustrate the ability of MMKG to efficiently represent facts that would be inexpressible by standard KGs; we also highlight the opportunities MMKGs offer to deal with complex tasks in standard KGs such as the entity alignment task.

- (1) *Different Modalities can bring out different semantics.* To illustrate this aspect, let us consider a community of Machine Learning researchers working on Convolutional Neural Networks,

who aim at staying abreast of the latest developments in CNNs. Different types of information could be useful to satisfy the needs of community members: some researchers may be interested in papers introducing novel algorithmic ideas in the field, while others may prefer to inspect source code - an implementation of the most recent research ideas. Finally, other researchers might be interested in accessing good-quality datasets to compare existing approaches. Here, papers, source codes, and datasets correspond to different modalities. It is difficult (and probably not useful) to compress all these modalities into a single one. The power of MMKGs emerges if we wish to support complex queries that involve independent modalities, such as “find me all papers about CNNs published in the last three years for which there is a PyTorch implementation on GitHub”. Managing this type of query with a standard KG would be very challenging, time-consuming, and perhaps impossible.

- (2) *Multi-modal Knowledge Graphs support Visual Query Answering [19]*. Suppose we have a photo depicting a famous monument, such as the Eiffel Tower or the Colosseum. A typical query would be to determine the location of the monument in the image, which is colloquially known as **Visual Query Answering (VQA)**. This task is relatively easy for humans who can identify objects in an image and associate them with their existing knowledge to correctly guess the location of a monument. However, the VQA task can be extremely challenging for AI agents. Standard KGs generally lack visual information that is essential for understanding an image. Although some efforts have been made to augment standard KGs with multi-modal knowledge, current approaches mainly represent knowledge through first-order predicates. An MMKG, on the other hand, should be able to use higher-order predicates to represent knowledge, making it more effective for tasks like VQA.
- (3) *Visual Data are easy to obtain and easy to verify*. As observed in [48], the visual representation of the real world is universal (think of an image depicting a cat or apple) and such a representation can be understood by every human; as a further advantage, images are often available for free on the Internet and they are of good quality. Finally, it is relatively easy to verify an image and to uniquely assign a meaning to each image.
- (4) *Visual data support the entity alignment tasks in KGs [43, 86]*. A crucial task in handling KGs is entity alignment, that is to decide whether two entities located in two independently constructed KGs refer to the same real-world object. Methods to align entities usually require inspecting the “neighborhood” of each entity in each KG and, if the degree of matching between extracted is “sufficiently good”, then the two entities identify the same object. However, entities could be described in different languages (e.g., English and Chinese); in some cases, given two entities E_1 and E_2 to match, it could happen that E_1 has a rich neighborhood while E_2 has a poor neighborhood. In all these cases, the entity alignment task is hard and even well-engineered algorithms fail to produce an accurate answer. In contrast, images are universal and, thus, if E_1 and E_2 are both equipped with visual information (i.e., an image containing only an instance of that entity), it would suffice to compare the associated visual information to decide whether E_1 and E_2 are aligned. To complete the discussion, we observe that the combination of visual and linguistic representation is effective in better identifying metaphors in texts [82] as well as in better computing if two words are related [104].

Existing scientific research clearly highlights the advantages of the multi-modal approach. For instance, in the biomedical field, [117] introduced a disease representation knowledge graph named **Specific Disease Knowledge Graph (SDKG-11)**. SDKG-11 is initially automatically constructed from biomedical literature and then combined with triples extracted from **Electronic Medical Records (EMRs)**, which have higher noise rates. However, the entity prediction experiments presented in [117] demonstrate that the multi-modal approach improves reasoning

accuracy, providing clinical practice descriptions closer to reality than scientific documents. Additionally, the integration of multiple modalities has been explored to reduce sparsity in recommender systems [89] and to detect fake news [96] and rumors [114] on social media.

4 DEFINITION, CONSTRUCTION, AND EXISTING MMKG

This section focuses on the definition and construction of MMKG and existing datasets. We first specify MMKG's definition and briefly explain its concept to tell people what it is (Section 4.1). Then, we introduce the construction methods of MMKG, which are vital because they show how to build MMKG data sets (Section 4.2). We also introduce the existing MMKGs and their situation (Section 4.3).

4.1 Definition of MMKG

The concept of a Multi-modal Knowledge Graph originates from the combination of multi-modal technology and structural information. The multi-modal technology was originally combined with the knowledge base, a system for representing knowledge, called **Multi-modal Knowledge** [3, 120]. The knowledge base is a broader concept than a knowledge graph. Later, the concept of a multi-modal knowledge graph was proposed by combining multi-modality and connectivity graphs. However, the connection graph cannot clearly represent the relationship between entities. Therefore, after the knowledge graph is widely used, multi-modality is more combined with the knowledge graph and becomes **Multi-modal Knowledge Graph (MMKG)**.

Regarding the definition of the MMKG, MKGAT model [89] firstly defines that MMKG introduces the information of other modalities into the traditional knowledge graphs. Some researchers [119] believe that the essence of multi-modal knowledge graph construction is associating symbolic knowledge, including entities, concepts, relationships, and so on, in traditional knowledge graphs with corresponding items. Similarly, we believe that the MMKG refers to a knowledge graph containing multiple types of information, such as text, images, audio, and so on, whose data has the same meaning but in different modalities will be represented uniformly. A Knowledge Graph KG can be formally defined as follows (see [119]):

Definition 4.1 (Knowledge Graph). A Knowledge Graph KG is a tuple $KG = \langle E, R, A, V, T_R, T_A \rangle$ where: (i) E is the set of entities, (ii) R is the set of relations, (iii) A is the set of attributes, (iv) V is the set of values that attribute can take on, (v) $T_R \subseteq E \times R \times E$ is the set of relation triples and (vi) $T_A \subseteq E \times A \times V$ is the set of attribute triples. More specifically, the triple $\langle e_1, r, e_2 \rangle \in T_R$ specifies that entities $e_1 \in E$ and $e_2 \in E$ are linked by relation $r \in R$, analogously, the triple $\langle e, a, v \rangle \in T_A$ indicates that the entity $e \in E$ has an attribute $a \in A$ which takes value $v \in V$.

A *Multi-modal Knowledge Graph* (hereafter shortened as *MMKG*) is a special case of a Knowledge Graph and, thus, its structure conforms to Definition 4.1. However, information stored in an MMKG can have different multimedia formats (called *modalities*) and, in detail, an MMKG might contain text snippets, images, videos, pieces of codes, and audio tracks which will be called *multi-modal data instances*.

The entities and relationships that populate a KG can be linked to multi-modal data instances in different ways. The definition of an MMKG needs to reflect how its entities/relationships are linked to multi-modal data instances, and this can be formalized as follows:

Definition 4.2 (Multi-modal Knowledge Graph). Let $M = \{m_1, m_2, \dots, m_p\}$ be a set of modalities and let $\mathcal{M}_i$ be the set of multi-modal data instances whose modality is m_i .³ Let $\mathcal{M} = \cup_{i=1}^p \mathcal{M}_i$ be

³For instance, if m_i is the modality corresponding to an image, then $\mathcal{M}_i$ will specify the set of all available images.

the set of all available multi-modal data instances. We define a *Multi-modal Knowledge Graph* K (or, in short, *MMKG*) one of the two following structures:

- (1) $K = \langle E, R, A, V, T_R, \rho(T_A, \mathcal{M}) \rangle$
- (2) $K = \langle E, R, A, V, \mu(T_R, \mathcal{M}), T_A \rangle$

Here ρ is an operator that extends the set of attribute triples to incorporate multi-modal data instances and μ is an operator that extends the set of relation triples to incorporate multi-modal data instances.

To make the Definition 4.2 more concrete, let us consider a practical example and suppose that our MMKG contains images as multi-modal data instances. A first option would be to consider an arbitrary image as an attribute associated with a specific entity, and in this case, the resulting MMKG would be called *A-MMKG*. In such a case, the ρ operator extends the set of attribute triples and, specifically, an attribute triple could consist of (i) an entity, an attribute, and the value such an attribute can assume or (ii) an entity, an attribute and a multi-modal data instance which replaces the attribute value considered in case (i). We can give the following definition:

Definition 4.3 (A-MMKG). Let $M = \{m_1, m_2, \dots, m_p\}$ be a set of modalities and let $\mathcal{M}_i$ be the set of multi-modal data instances whose modality is m_i . Let $\mathcal{M} = \cup_{i=1}^p \mathcal{M}_i$ be the set of all available multi-modal data instances; finally, let $m_i \in \mathcal{M}$ be an arbitrary element in $\mathcal{M}$.

An A-MMKG K is a tuple $K = \langle E, R, A, V, T_R, T'_A \rangle$ where the symbols E, R, A, V , and T_R have the meaning we provided in Definition 4.1. The set of attribute values T'_A is defined as $T'_A \subseteq E \times A \times (V \cup \mathcal{M})$, that is an attribute triple associates an entity $e \in E$ with an attribute $a \in A$ and a value $v \in V$ or, alternatively, a multi-modal data instance $m_i \in \mathcal{M}$.

A different strategy consists to consider each multi-modal data instance as a *stand-alone entity* and the corresponding MMKG is called *N-MMKG*. In such a case the μ operator extends the set of relation triples because the head of a relation triple could be an entity or a multi-modal data instance and, simultaneously, the tail of the relation could be another entity or another multi-modal data instance. We can give the following definition:

Definition 4.4 (N-MMKG). Let $M = \{m_1, m_2, \dots, m_p\}$ be a set of modalities and let $\mathcal{M}_i$ be the set of multi-modal data instances whose modality is m_i . Let $\mathcal{M} = \cup_{i=1}^p \mathcal{M}_i$ be the set of all available multi-modal data instances; finally, let $m_i \in \mathcal{M}$ and m'_i be two arbitrary elements in $\mathcal{M}$.

A N-MMKG K is a tuple $K = \langle E, R, A, V, T'_R, T_A \rangle$ where the symbols E, R, A, V , and T_A have the meaning we provided in Definition 4.1. The set of relation triples T'_R is defined as $T'_R \subseteq (E \times \mathcal{M}) \times A \times (E \times \mathcal{M})$, that is relation triple r associates an entity $e_1 \in E$ or a multi-modal data instance $m_i \in \mathcal{M}$ with another entity $e_2 \in E$ or another multi-modal data instance $m'_i \in \mathcal{M}$.

A graphical representation of A-MMKG and N-MMKG strategies is reported in Figure 3. Here we considered a fragment of the Richpedia knowledge base; the same fragment has been encoded according to the A-MMKG strategy (see the left side in Figure 3) and the N-MMKG strategy (see the right side of Figure 3). To gain a better understanding of the difference between A-MMKG and N-MMKG, let us focus on an MMKG representing cities in the world and assume that it contains the entity “Rome”. If we would opt for the A-MMKG strategy, then the entity “Rome” would have attributes such as “Italy” as well as images depicting the city of Rome. In contrast, if we would opt for the N-MMKG method, then all images about the city of Rome have to be considered as independent entities belonging to the same KG.

In addition to categorizing MMKG based on whether the attributes in MMKG are considered separate entities, we can also categorize MMKG into Global-MMKG and Local-MMKG based on

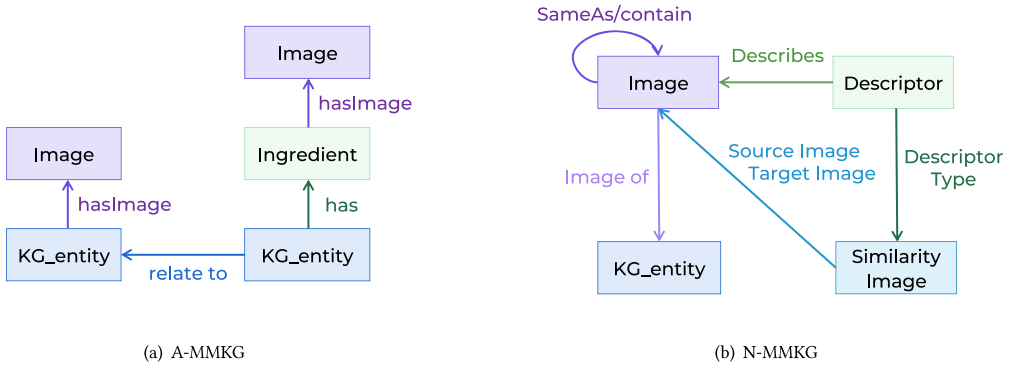


Fig. 3. A MMKG drawn from RichPedia. In the left side, we applied the A-MMKG strategy while in the right side we applied the N-MMKG strategy.

whether the knowledge represented by the knowledge graph can be applied globally. Here, Global-MMKG refers to knowledge graphs with extensive content that spans multiple images, such as Richpedia and MMKG, while Local-MMKG refers to knowledge graphs extracted from specific text and images, such as some event diagrams extracted by the CLIP-Event model. Compared to Global-MMKG, the scope of Local-MMKG is limited, but it can help with understanding and reasoning about information. Currently, all MMKG datasets belong to the category of Global-MMKG.

We can provide a further categorization that specifies how entities and modal data (in particular, images) are related. In the first case (see for instance Richpedia [93]), there is a one-to-many relationship between an entity and a collection of images: as an example, a popular monument such as *The Big Ben* might be associated with several images depicting it. In this case, we call our MMKG as a *Global-MMKG*. In contrast, suppose (as in the CLIP-EVENT approach [40]) that there is a one-to-many relationship between an image and a collection of entities: for example, an image could contain the faces of several famous people and, thus, we could associate such an image with multiple entities, each of them corresponding to a famous person. In this case, we call our MMKG as a *Local-MMKG*. More formally, we give the following definition:

Definition 4.5 (Global-MMKG and Local-MMKG). Let K be a multi-modal knowledge graph in which E is the set of entities and I is the set of available images (or videos, audio clips, and so on). We say that K is a *Global-MMKG* if there exists a function $\phi : E \rightarrow P(I)$ that maps an entity $e \in E$ to a subset $I' \subseteq I$ of images in I . Here $P(I)$ is the power-set of I . We say that K is a *Local-MMKG* if there exists a function $\psi : I \rightarrow P(E)$ that maps an image $i \in I$ to a subset $E' \subseteq E$ of entities in E . Here $P(E)$ is the power-set of E .

4.2 Methods to Construct MMKG

This section focuses on the construction methods of MMKGs, that is, Multi-Modal Knowledge Graphs. Specifically (see Section 4.2.1), we first introduce a high-level conceptual framework that identifies the main steps of the process to construct an MMKG.

Subsequently, Section 4.2.2 provides a brief overview of the methods to construct an MMKG if input data are assumed to be *structured* while Section 4.2.3 deals with methods to construct MMKGs from unstructured data sources. In both the case of structured and unstructured data sources, we explain how a specific method implements the various steps portrayed in Section 4.2.1. There is no difference in superiority or inferiority among different construction methods. They are just construction methods that are suited for the different data types. In our opinion, our conceptual

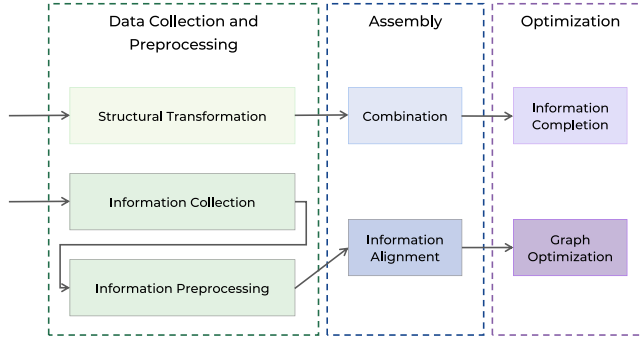


Fig. 4. Workflow of construction methods.

framework makes it easier to grasp the similarities and differences between seemingly very different methods as well as to clarify the pros and cons of a specific construction method over the other ones.

4.2.1 Generic Construction Framework. In this section, we introduce a conceptual framework to model the construction of an MMKG. The framework includes three steps: *data collection and preprocessing*, *assembly*, and *optimization*. Figure 4 pictorially reports the whole workflow to construct an MMKG from both structured and unstructured data sources.

Data Collection and Preprocessing. The first step to constructing an MMKG is to collect data and pre-process it to transform it into a suitable format. Some approaches to data collection gather raw data by crawling the Web, while other approaches directly query existing databases. The data preprocessing on the collected information is performed to transform collected data into a suitable format. This step responds to the structural transformation in construction based on structured data and the information collection and information alignment in construction based on unstructured data.

Assembly. After performing the data collection and preprocessing step, existing approaches assemble collected data into an MMKG. Combining structured data during construction and aligning information in unstructured data during construction both belong to the assembly step.

Optimization. The third step in constructing an MMKG involves refining and optimizing the MMKG created in the first two steps, making it more widely applicable to various downstream tasks. This step includes information completion for structured data and graph optimization for unstructured data. In addition, text and visual data are often integrated to support advanced queries, with RDF links used to connect entity descriptions to corresponding images. This step aims at enhancing the MMKG’s ability to handle complex queries and effectively represent knowledge.

4.2.2 Construction Based on Structured Data. We are now able to discuss how the general framework introduced in Section 4.2.1 specializes to structured datasets. The framework consists of three main stages, that is, (i) *structural transformation stage*, (ii) *combination stage*, and (iii) *information completion*.

Structural Transformation. Most of the structured data sources are relevant to build up an MMKG in databases. For example, Zhu et al. [85] describes an approach to building a large-scale multi-modal KB to answer the visual query (that is, a query like “*find me pictures containing a cat*”) by evaluating the probability KB query. The authors consider the SUN dataset and, after fetching the data, the model needs to perform a triple transformation on the structured data. The model above converts large-scale **Markov Random Fields (MRFs)** to KB representations by performing

a data preprocessing step followed by a factor graph generation stage. Specifically, the approach of [85] constructs a factor graph for each image, connecting variables to values in the database. The model uses a declarative language developed by the authors to reconnect the relationships between multi-modal entities and the Gibbs sampler to learn the factor weights in factor graphs. The generation of the AMPLUS dataset [15] relies on converting the obtained XML files into rough RDF and reorganizing the data using interactive rewriting rules within the XMLRDF tool. The advantage of this approach is that implicit links can also be explicitly represented as RDF relationships.

Combination. After structural transformations, it's necessary to integrate various knowledge graph and modal information to construct MMKGs. Liu et al. [50] outlined a method for MMKG construction: aligning most entities from FB15K, a popular dataset for link prediction algorithm validation, with entities from two other popular knowledge graphs, namely DBPEDIA and YAGO, using the *sameAs* relationship, resulting in DBPEDIA15k and YAGO15k. The format of the triples is *(head?, sameAs, tail)*, where the head and tail entities are from different knowledge graphs. The MARS model [115] proposed by Zhang merges relationships and entities with the same meaning when constructing MarKG and MARS. After the merging process, the model obtains 38 types of relationships, which can make MarKG and MARS more in line with the requirements for the model's reasoning and evolution. In the field of cultural heritage, **European Data Model (EDM)** serves as an open cross-disciplinary semantic web that accommodates rich metadata. The implementation of Amsterdam Museum Linked Data dataset [15] with EDM interoperability is an instance of combination. This system maps specific properties to the attributes in the metadata schema, allowing for interoperability while retaining the complexity of the original data.

Information Completion. However, the existing structured data sources relevant to constructing an MMKG are small, and thus the ability to construct an MMKG through operations like the combinations we described before is limited. In recent years, researchers have studied appropriate strategies for integrating KGs derived from texts with other modalities. To this end, web crawlers, along with ad-hoc search engines, are promising technologies for properly enriching a KG that is derived from textual facts. For instance, the FB15K knowledge graph does not include visual modality information. To address this issue, [67] queried search engines like GoogleImage to retrieve images associated with entities. More specifically, every entity in FB15K is linked to the first 25 images returned by the search engine. ImageGraph [67] is an MMKG that extends the FB15K dataset we cited before; it contains 1,330 relation types, 14,870 entities, and 829,931 images crawled from the Web. As far as we know, ImageGraph represents a pioneering effort in utilizing visual data linked to an entity as query parameters. The incorporation of visual data unlocks the potential for various innovative query types, including: (i) *Discovering unknown relations between entities by analyzing pairs of unfamiliar images whose corresponding entities are unknown* and (ii) *Completing queries by identifying the seen images that match an unseen image and an unknown KG entity, along with a specified relation type*. Web crawling methods have been also applied to the construction of Richpedia [93]. Richpedia collects entities and images from four image search engines, that is Wikipedia,⁴ Google,⁵ Bing,⁶ and Yahoo.⁷ Each image is stored in Richpedia as an entity. After this, RDF links are established between image entities based on the hyperlinks and descriptions available from Wikipedia. One of the most innovative features of RichPedia is that entities derived from textual information are uniquely identified. Similarly, the MARS model also uses web crawlers to collect images to supplement its knowledge graph information. The model collects images from two sources, Google Engine, and Laion-5B query service.

⁴<https://www.wikipedia.org>

⁵<https://www.google.com>

⁶<http://www.bing.com>

⁷<https://www.yahoo.com>

4.2.3 Construction based on Unstructured Data. The task of constructing an MMKG from one or more unstructured data sources is much more challenging than the case of structured data sources. Facing unstructured data, models must process data from scratch, such as information extraction or unimodal representation mentioned in Section 2. We consider four main steps in the task of constructing an MMKG, that are information collection, information preprocessing, information alignment, and graph optimization.

Information Collection. The information collection step relies on search engines or other tools to extract information. **Never Ending Image Learner (NEIL)** [13] collects visual information from Internet data runs 24/7 by Google Image Search; it applies a semi-supervised learning algorithm that finds out common sense relationships (for instance *wheel is part of a car*) and it labels instances of the given visual categories. The MCTKG system [101] applies regular expressions to obtain semi-structured data from major tourism websites including the Beijing tourism website, Baidu Encyclopedia, and so on. Kannan et al. describe the **Deep Curator System (DCC)** system [37], which attempts at extracting information from scientific publications in the field of Deep Learning and the accompanying source code and mapping extracted information on a unified KG. Here the authors develop modules to process three types of modalities, that is text, images, and code and they describe an approach to aligning collected information into a unique multi-modal KG.

Information Preprocessing. Information preprocessing refers to cleaning the collected information's redundant, missing, and faulty parts and converting them into higher-quality information. NEIL conducts image understanding and object/scene recognition on raw data to extract more text information and relationships. It uses an iterative method in cleaning labels and uses detectors and classifiers to learn relationships based on statistics. The MCTKG applies a semi-automatic approach to construct a global ontology to catch the main entities (along with their hierarchical relationships) in the tourism domain.

Information Alignment. Information alignment mainly normalizes synonymous information. Traditional coreference and disambiguation methods can be used for synonymous text information alignment. Synonymous information of other modalities needs to be aligned by representation or fusion of MMKG. The MCTKG system provides an entity alignment algorithm based on N-gram distance. The DCC systems apply a two-stage strategy to perform entity alignment: in the first stage (called *local alignment*), each modality instance is mapped onto an element of a conceptual schema; in the second stage (called *global alignment*), DCC makes use of an external curated KG called **Computer Science Ontology (CSO)** which has been extensively used to categorize scientific articles in Computer Science.

Graph optimization. Completion and optimization of the graph generally rely on the Crawler mentioned in the construction method based on structured data or use the fusion of MMKG to predict and supply new relationships. MCTKG copies attribute value to the attribute lacking entity for information completion.

4.3 Existing MMKGs

In this section, we provide an overview of the MMKG datasets, including the number of entities, triples, relationships, and images that they comprise. Additionally, we highlight two key factors that influence the size and quality of MMKG datasets: the construction methods employed and the type of MMKG.

4.3.1 Features of Existing MMKG. Table 2 provides an overview of the most popular MMKGs along with some statistics about them (such as the number of entities, triples, and the number of distinct types of relationships stored in each MMKG). Over the years, there has been a substantial

Table 2. Some Statistics about Existing MMKGs

Model	Entities	Triples	Relationships	Image	Type
NEIL [13]	2,186	–	1,703	2M+	N-MMKG
KB [120]	–	–	–	–	N-MMKG
ImageGraph [67]	14,870	564,010	1,330	829,931	A-MMKG
DB15k (MMKG) [50]	14,777	170,836	279	12,841	A-MMKG
Yago15k (MMKG) [50]	15,283	193,684	32	11,194	A-MMKG
Richpedia [93]	2,944,755	119,669,570	–	2,914,770	N-MMKG
GAKG [16]	8,991,737	68,629,515	41,664,304	–	A-MMKG
DCC [37]	7,999	–	–	–	–
MCTKG [101]	47,718	390,746	–	–	–
K3M [121]	40,132,418	288,646,296	–	–	N-MMKG
Alime MKG [106]	300K+	514K+	–	28M+	A-MMKG
MarKG [115]	11,292	34,420	192	76,424	N-MMKG
MARS [115]	2,063	–	27	13,398	N-MMKG
amplus [15]	1,026,162	2,521,046	33	56,130	N-MMKG
DMG [98]	262,494	1,850,451	62	58,791	A-MMKG
mdgenre	191,135	1,252,247	154	67,528	–

Here M stands for millions while K denotes thousands.

increase in the size and complexity of available MMKGs. Moreover, the scope of MMKGs has expanded beyond encyclopedic knowledge to encompass a diverse range of fields, including academics, tourism, scientific literature, and e-commerce. The rapid growth in the number of available MMKGs, along with their increasing complexity, are clear indicators of their success in a wide range of domains. However, some non-trivial challenges arise that need to be properly addressed. Specifically, the sizes of MMKGs may differ by some orders of magnitude, and the task of combining two or more MMKGs to generate a bigger one could be computationally demanding. Furthermore, due to commercial confidentiality, several large-scale MMKG datasets that generate profit are not publicly accessible, making their utilization and re-use impossible.

4.3.2 Factors Impact MMKG. This research identified two factors that significantly impact the situation of MMKG datasets, that is, *construction methods* and *MMKG type*.

Construction Methods Factor. Different construction methods deeply affect the size of an MMKG. Structured data construction methods, in fact, tend to produce smaller datasets than construction methods based on unstructured data sources; such behavior depends on the fact that much of the information that cannot be aligned with existing entities is discarded during the combination process. For instance, the DB15K and YAGO15K datasets synthesized in [50] contain approximately 10,000 entities and 100,000 triples. The MarKG and MARS datasets synthesized in [115] contain no more than 12,000 entities and 35,000 triples.

In contrast, unstructured data construction methods generate datasets with a larger number of entities and triples. Richpedia and GAKG, for instance, comprise over one million of entities and 100 million triples. Such a behavior depends on the fact that methods based on unstructured data often use resources such as Web pages returned by search engines, which typically provide much more information than existing knowledge graphs. This explains why MMKGs obtained from unstructured data sources are generally much bigger than datasets obtained from structured data sources.

MMKG Type Factor. The second factor we wish to consider is the type of an MMKG and, to this end, we observe that the quality of an N-MMKG may significantly differ from the quality of an A-MMKG. As demonstrated by EVA [48] and ImageGraph [67], N-MMKGs offer the advantage of

dense information. In other words, in N-MMKGs, we link an image with multiple entities, and thus an image serves as a signal to align entities located in different KGs, often in different languages. The capability of images to act as a bridge entity is particularly appreciable in case of *rare entities* without adequate contextual information (see [10, 27]): as observed in [48], in fact, it is very hard to match rare entities if we would only rely upon textual descriptions or the neighbors of each entity in each KG while such a task becomes relatively easy if an image is associated with entities to align. However, N-MMKGs also contain relatively mixed information, necessitating complex data preprocessing.

In contrast, A-MMKGs, exemplified by Richpedia, provide explicit information, making the preprocessing step more straightforward, and have higher efficiency and accuracy in alignment. Nevertheless, A-MMKGs fail to conform to the information distribution on the Internet, requiring more work in information completion, such as crawling. Furthermore, A-MMKGs cannot reflect image information with multiple entities and relationships, resulting in a reduction and loss of data.

5 REASONING OVER MMKG

Since reasoning over KG is a fundamental part of reasoning over MMKG, we will first introduce reasoning over KG before discussing reasoning over MMKG. *Knowledge Graph Reasoning* (often abbreviated as KGR) is an essential tool to expand a Knowledge Graph KG by inferring new facts from the facts which are currently stored in a KG. We refer interested readers to a very recent and very detailed survey on KGR [44] and, for convenience, we provide below a formal definition of KGR:

Definition 5.1. Let $KG = \langle E, R, A, V, T_R, T_A \rangle$ be a Knowledge Graph which conforms to Definition 4.1: the Knowledge Graph Reasoning (KGR, for short) aims at generating a new triple $t^* = \langle e_1^*, r^*, e_2^* \rangle \notin T_R$ which is associated with likelihood $f(t^*)$. The KGR activity can be further classified in: (a) *head reasoning*, if we want to predict e_1^* given that $r^* \in R$ and $e_2^* \in E$, (b) *tail inferring*, if we want to predict e_2^* given that $e_1^* \in E$ and $r^* \in R$ and (c) *relation inferring*, if we want to predict r^* given that $e_1^* \in E$ and $e_2^* \in E$.

Definition 5.1 easily extends to temporal knowledge graphs, which are defined as a sequence $S_t = \{KG_1, KG_2, \dots, KG_n\}$ of knowledge graphs, where the ℓ -th element of the sequence (with $1 \leq \ell \leq n$) is the knowledge graph KG_ℓ . If we manage temporal knowledge graphs, the goal of KGR is to predict a quadruple of the form $\langle e_1^*, r^*, e_2^*, t^* \rangle$. Finally, KGR can be defined for multi-modal knowledge graphs, and, in this case, we assumed that both the entities and relations in the KG are endowed with multiple modalities that can be combined to make more accurate inference.

In order to perform KGR effectively, we need an accurate KG construction phase, and furthermore it is also necessary to have an effective representation of a KG. Consequently, two techniques are at the basis of successful reasoning about a KG, namely *alignment* (i.e., the ability to recognize that entities/relations from different sources and with different names represent the same real-world object/relation and therefore need to be treated as a unique object) and *embedding* (i.e., the ability to represent semantically heterogeneous entities and relations as points in a single vector space). In the following, we will consider alignment and embedding as key components of the KGR process and, in line with previous work in multi-modal machine learning (see [4] for a detailed review), we will refer to these techniques as *challenges*. This section includes *representation*, *fusion*, *alignment*, and *translation*, that can actually contribute to improving the quality and richness of an existing MMKG. We describe how existing approaches address one or more of these challenges.

Pre-training techniques are becoming increasingly popular for reasoning over knowledge graphs, and especially for multi-modal knowledge graphs. Because pre-training techniques have been successfully used to improve knowledge representation, they should be considered as a powerful form of KG embedding. However, considering pre-training techniques solely as a technique

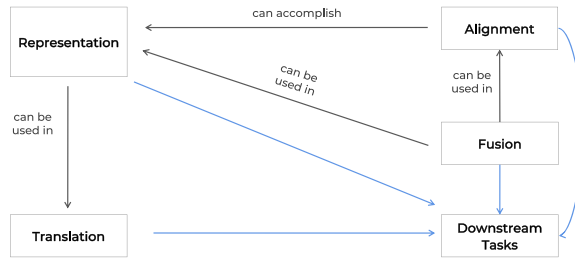


Fig. 5. Connections among four fundamental challenges of MMKG.

to pursue KG embedding is, in our view, too reductive, as pre-training techniques can be applied to solve a wide range of downstream tasks, such as question generation in natural language and noise correction in a KG. Because of its importance (which we firmly believe will increase in the next years), we discuss pre-training in multi-modal knowledge graphs in detail. This section covers feature embedding pre-training, auxiliary pre-training, and downstream task pre-training, emphasizing the versatility of pre-training techniques and highlighting their supportive role in reasoning tasks. Our classification approach is *flexible*, i.e., it allows us to easily catalog future approaches.

5.1 Four Fundamental Challenges of MMKG

In this section, we present four fundamental challenges that are relevant to expanding the knowledge and improving the accuracy of an MMKG, partly derived from multi-modal machine learning. We introduce the close connection between these challenges and discuss them one by one.

5.1.1 Connection among Four Fundamental Challenges. Through our research, we have identified that the four fundamental challenges in MMKG are interconnected and work in a synergic fashion to enhance the representation of knowledge. Completing a downstream task or basic challenge with an MMKG model may involve one or more underlying challenges, as Figure 5 shown. For instance, alignment and fusion techniques can be used together to achieve better representation. Fusion techniques such as early, late, and hybrid fusion can be applied to aid in alignment. Precise representation is crucial in enabling modal translations in MMKG, which is essential for downstream tasks. The diagram below illustrates the relationships between the challenges, with black arrows indicating the interconnections and blue arrows highlighting their relevance to downstream tasks. It is important to note that due to the close interdependence of these challenges, their concepts may be easily confused, and their boundaries should be more explicit. Therefore, we will emphasize their definitions in MMKG and distinctions in each subsection.

5.1.2 Representation. Representation learning is an essential component of any machine learning task. In short, it consists of transforming data from various sources into high-dimensional vectors, often called *embeddings*. Representation learning in MMKG is particularly challenging because, on one hand, embeddings should capture the structural information of a knowledge graph (so, for instance, if two entities are directly tied by a relationship, if they are tied by a multi-hop path or, finally, if no path joining them exists) along with multi-modal instances (so, for instance, if an entity is associated with text snippets, one or more images and so on). A comprehensive embedding of entities and relationships in an MMKG enables machines to better understand the relationships between entities and their multi-modal information, allowing for more advanced reasoning capabilities.

Representation learning in MMKG can be classified into two categories, namely: *joint representation* and *coordinated representation*. Joint representation maps all multi-modal information onto one high-dimensional embedding, while coordinated representation extracts hidden information

from each modality and reflects it into a shared space under specific constraints. However, real-world datasets often contain noisy data, and they are inherently incomplete. Due to these factors, the representation learning for MMKG is challenging, and potential inaccuracies in the construction of entity/relationship embeddings can have negative effects on downstream tasks. Given these premises, we require that the representation learning process in MMKG must consider the consistency and complementarity of multi-modal information while also eliminating redundancies.

Colloquially speaking, we identify two main components in the representation learning task: the first component is about the neural network architecture we adopt to generate embeddings; the second component is about the scoring and loss function we exploit to perform learning tasks. In principle, there are many neural network architectures that can be used in training. However, the key point is to choose a scoring/loss function that makes the representation more accurate and meaningful. In the following, we list models on representation in Table 3 and describe how they address the above problem.

Neural Networks. MMKG models use various neural networks to extract hidden information and embed modal information. These neural networks are designed for different types of data, including text, numerical, visual, and structured data, and they are able to produce high-quality embeddings.

For structured data such as knowledge graphs, MMKG models typically employ **Fully Connected Networks (FCNs)** to generate vectors that represent the entities and their relationships. Specifically, MMKG models generally use sequential networks (e.g., biGRU in MKBE [71] and LSTM in MKGAT [89]) to handle variable-length text data. Sequential networks can effectively capture sequential dependencies from texts and produce meaningful embeddings.

Another commonly used network for performing graph structure embedding in knowledge graphs is GCN. GCN is a type of deep learning model based on graph-structured data. Compared to FCN, GCN can consider neighboring information of nodes, utilize richer local information, and produce more accurate representations. Additionally, GCN can aggregate information at different depths, enabling effective learning of local and global information for nodes of different scopes.

Visual information is also an important modality in MMKG models. VGG [83] and ResNet [30] are popular choices to embed visual information because they achieve high performance without requiring extensive training. These networks are able to extract high-level features from images, which are then used to generate embeddings for the entities in the knowledge graph.

Numerical information is another important modality in MMKG models. Feedforward layers or algorithms based on them, such as RBF [9], are commonly used to embed numerical information. These layers are able to capture the non-linear relationships between numerical features and generate embeddings that can be used to reason about the relationships between entities.

Scoring Function. As already observed, a crucial aspect of MMKG representation is the choice of scoring and loss functions. In contrast to traditional knowledge graphs, which are usually represented as directed graphs of entities and their relationships, MMKG also incorporates additional modalities, such as images, videos, and text. Therefore, the selected scoring function should be capable of effectively capturing the complex relationships between different modalities and entities.

The selection of a scoring function for MMKG construction depends on specific requirements. For example, the MKBE model uses ConvE and DistMult without negative samples. ConvE is suitable for modeling complex relationships between entities and relations, while DistMult is effective at modeling interactions between different modalities. In contrast, MKHAN uses TransH to model a relation as a hyperplane because it is proficient in capturing various relation semantics, particularly asymmetric relations.

Similarly, MKGAT and MMEA employ TransE due to their simplicity and computational efficiency, making it an excellent choice for large-scale knowledge graphs. It is important to note that

Table 3. Comparison of Some Approaches to Constructing an MMKG on the Basis of the Representation Learning Technology they Adopt

Model	MKBE [71]	MKGAT [89]	MKHAN [116]	MMEA [12]	MMKG [50]	EVA [48]
S-Info Representation	Dense Network	Embedding Network	FCN	<i>not mentioned</i>	PoE	GCN
T-Info Representation	biGRU	Word2Vec + SIF/LSTM	–	<i>not mentioned</i>	–	FFN
V-Info Representation	VGG	ResNet50	ResNet50	VGG16	VGG16	ResNet152
Num-Info Representation	FFN	–	–	RBF	PoE	–
Scoring Function	ConvE/ DistMult	TransE	TransH	TransE	PoE	–
Model Mission	Representation	Recommendation	Route Representation	Entity Alignmnet	Dataset Creation	Entity Alignment
Dataset	WN9-IMG [103]	MovieLens [28]/ DianPing ⁸	ChunYu ⁹ / DianPing	DB15K/ YAGO15K [50]	FB ¹⁰ /DB ¹¹ / YAGO [88]	DB15K [50]/ DWY15K [25]
Info-Types	S+V	S+L+V	S+L+V	S+L+V	S+L+V	S+L+V
Model	DFMKE [118]	MKRLN [91]	ImageGraph [67]	DCC [37]	K3M [121]	Serigieh [62]
S-Info Representation	GCN	GCN	Embedding Network	GCN	GCN	<i>not mentioned</i>
T-Info Representation	–	GCN	–	–	GCN	Word2Vec/ AutoExtend
V-Info Representation	ResNet	VGG19	VGG16	ResNet	VGG19	VGG19/ VGG128
Num-Info Representation	FFN	GCN	–	FFN	GCN	–
Scoring Function	TransE	TransE	TransE/ DistMult/ HOLE/ concat	TransE	TransE	TransE
Model Mission	Entity Alignment	Recommendation	Question Answering	Entity Alignment	Recommendation	Representation
Dataset	DB15K/ YAGO15k	Movie/Book/ KKBOX ¹²	ImageGraph	DB15K/ DWY15K [25]	DB15K/ YAGO15K	WN9-IMG/ FB-IMG
Info-Types	S+L+V	S+L+V	S+V	S+L+V	S+L+V	S+L+V

Here, the horizontal bar means the model lacks of that modal information. S-Info and S mean structural information. T-Info means Textual information. V-info and V mean visual information. Num-Info means numerical information. L means linguistic information, including textual and numerical information.

the scoring function's effectiveness depends on its alignment with the data's characteristics. Therefore, selecting the appropriate scoring function requires a good understanding of the MMKG's characteristics to be modeled.

5.1.3 Fusion. Fusion methods integrate different modalities of information into a single, unified representation vector with the purpose of producing a more detailed and precise representation of entities and relationships. Due to these reasons, fusion methods have been successfully applied in a wide range of applications, from image and speech recognition to natural language processing. In the case of MMKGs, fusion techniques target integrating information from multiple modalities. There are three main types of MMKG fusion techniques, that is early fusion, late fusion, and hybrid fusion. In the following, we describe each of them in detail. Table 4 presents several approaches that

⁸<https://www.dianping.com>

⁹<https://www.chunyuyisheng.com/pc/qalist>

¹⁰<http://www.freebase.com>

¹¹<https://www.dbpedia.org>

¹²<https://www.kaggle.com/c/kkbox-music-recommendation-challenge/data>

Table 4. Some Approaches that Implement Fusion Methods for Constructing and Managing an MMKG

Fusion Type	Model	Method	Mission	Dataset
Early Fusion	MKGAT [89]	Dense Layer	Fake News Detection	MovieLens/DianPing
Late Fusion	MKBE [71]	Scoring Function	Representation	WN9-IMG
	MKHAN [116]	Energy Function	Question Answering	ChunYu/DianPing
	MMEA [12]	Energy Function	Entity Alignment	DB15K/YAGO15K
	MMKG [50]	PoE	Dataset Creation	FB/DB/YAGO
	EVA [48]	Attention Weighted Concatenation	Alignment	DB15K/DWY15K
	Sergieh [62]	Energy Function	Representation	WN9-IMG/FB-IMG
	K3M [121]	Initial-interactive Feature Fusion	Dataset Creation	YAGO15K/DB15K
Hybrid Fusion	DFMKE [118]	concatenation/LMF	Entity Alignment	DB15K/YAGO15K

implement fusion strategies for constructing an MMKG. For each approach, we provide details on the type of fusion strategy employed, along with the fusion method used. Additionally, we include information on the mission of each approach and the dataset on which it has been tested, all within the same table.

Early Fusion. In early fusion strategies, features are combined immediately after they are extracted, allowing them to be analyzed together from the outset. Early fusion helps to increase the amount of available information and improve the accuracy of the analysis especially when we need to deal with small datasets. Early fusion techniques offer two main advantages: they are capable of quickly generating an output and they help reduce overfitting. In fact, since features are combined at an early stage, the processing time required for each modality is reduced. Early fusion aids in mitigating the likelihood of overfitting, which arises when a model is excessively fine-tuned to a particular dataset, resulting in subpar generalization performance on new and unseen data.

Only the MKGAT system implements early fusion methods. In detail, it employs a simple dense layer to learn a representation from the information associated with each modality. By fusing the single representation, it obtains a dense vector associated with the entity. The dense layer allows the intrinsic weight to be adjusted through learning, resulting in a more suitable entity embedding. This approach is effective in cases where the modalities have different data distributions and can extract distinct features.

However, using dense layer to complete early fusion suffers from some limitations. Firstly, early fusion methods require fixed input size for each modality, but such an assumption fails to be true in some real datasets. Secondly, early fusion methods may not be able to capture the complex inter-modal relationships between different modalities, which can limit the model's performance.

Late Fusion. Late fusion is a family of techniques in which features extracted from different sources and/or associated with different modalities are first processed independently. After this pre-processing step finishes, processed features are combined to get a more detailed representation of information. Late fusion methods are frequently employed when working with large datasets or complex problems since they allow for a more sophisticated and nuanced understanding of the data.

One advantage of late fusion methods is that they can increase the weight of the shared hidden information extracted from each modalities and such an integration can provide complementary information that might not be present in any one modality alone. Another benefit of late fusion

methods is that they can handle missing data or incomplete features. For instance, in some scenarios, some features might be missing from one modality but present in another. By combining the features from different modalities, late fusion can effectively handle missing data and improve the overall performance of the system.

As emerges from Table 4, late fusion are the most commonly used fusion method, with energy function, **Products of Experts (PoE)**, **Low-rank Multi-modal Fusion model (LMF)/Tensor Fusion Network (TFN)**, and other self-setting modules being the fusion ways utilized to fuse features. The minimization of a suitable energy function is the most widely used technique to perform late fusion in MMKGs. We recall that energy functions were used by TransE [7] for the first time: more specifically, given a triple (h, r, t) , let h_s, r_s and t_s be the structural embedding of h, r and s respectively. The (structural) energy function E_S is defined as follows:

$$E_S = ||h_s + r_s - t_s|| \quad (1)$$

Our goal is to modify the definition of E_S by adding further terms which reflect the contribution of all available modalities.

The first attempt at generalizing the definition of E_S to incorporate multi-modal features due to Mousselly–Sergieh et al. [62] who provided the following definition:

$$E = E_S + E_{MS} + E_{SM} + E_{M1} + E_{M2} = ||h_s + r_s - t_s|| + ||h_m + r_s - t_s|| + ||h_s + r_s - t_m|| + ||h_m + r_s - t_m|| + |(h_m + h_s) + r_s - (t_m + t_s)| \quad (2)$$

To explain this modified definition of energy, let h_s and t_s represent the structural embeddings of the head and tail, respectively. Similarly, we denote h_w/t_w and h_i/t_i as the linguistic and visual embeddings of the head and tail. Next, we concatenate the linguistic and visual embeddings to obtain the multi-modal embeddings h_m and t_m . In the framework above, the contribution E_{M1} implies that the multi-modal embeddings of the head and tail, once projected into the structural space, should be as close as possible. The energy contribution E_{M2} models the constraint that if we sum the structural and multi-modal representations of both the head and tail, the resulting vectors should be as close as possible. Finally, the two terms E_{SM} and E_{MS} enforce the structural and multi-modal embeddings of both the head and tail to reside in the same space.

The MKHAN system applies a similar definition of energy; specifically, with the notation above, let us consider the following transformations:

$$h_{\perp}^s = h_s - w_r^T h_s w_r \quad \text{and} \quad t_{\perp}^s = t_s - w_r^T t_s w_r,$$

and

$$h_{\perp}^m = h_m - w_r^T h_m w_r \quad \text{and} \quad t_{\perp}^m = t_m - w_r^T t_m w_r,$$

with $||w_r||_2 = 1$. Finally, MKHAN defines the projection of combined representations as follows:

$$h_{\perp}^C = h_{\perp}^s + h_{\perp}^m \quad \text{and} \quad t_{\perp}^C = t_{\perp}^s + t_{\perp}^m.$$

With the premises above, the energy function is MKHAN reads as follows:

$$E = E_S + E_M + E_C + E_{SM} + E_{MS} = ||h_{\perp}^S + d_r - t_{\perp}^S|| + ||h_{\perp}^M + d_r - t_{\perp}^M|| + ||h_{\perp}^C + d_r - t_{\perp}^C|| + ||h_{\perp}^S + d_r - t_{\perp}^M|| + ||h_{\perp}^M + d_r - t_{\perp}^S||. \quad (3)$$

In other words, the energy function is the sum of three separate contributions, that is a structural contribution E_S , a multi-modal contribution E_M , and a combined contribution E_C ; in addition, two further contributions, namely E_{SM} and E_{MS} are considered and they have the same meaning of the E_{SM} and E_{MS} contribution introduced [62]. A further interesting approach to fusing features is due to the Multi-Modal Entity Alignment (MMEA, in short) framework [12]. Unlike previous approaches, MMEA separately considers entity embedded in the relational (E_r), visual (E_i), and numerical (E_n) spaces and it attempts at finding an embedding in a common space. If we let E be

the entity embedding in the common space, then E is obtained by minimizing the following loss function:

$$L_{csl}(E, E^{(r)}, E^{(i)}, E^{(n)}) = \alpha_1 \|E - E^{(r)}\|_2^2 + \alpha_2 \|E - E^{(i)}\|_2^2 + \alpha_3 \|E - E^{(n)}\|_2, \quad (4)$$

in which α_1 , α_2 and α_3 are suitable hyperparameters.

Products of experts. (PoE) is a powerful probabilistic modeling technique that combines the outputs of multiple expert models to capture complex relationships between features. Each expert model is trained to capture the statistical relationships between a subset of features in the dataset. The output of each expert model is then combined using a product or weighted sum, resulting in the overall output of the PoE model. PoE models are particularly effective at capturing complex, nonlinear relationships between variables that may be difficult to model using other techniques. This makes PoE a popular choice in a range of applications, from image and speech recognition to natural language processing and reinforcement learning. To describe how PoE works, let us consider the embedding expert $f_{(r,L)}(d|\theta_{(r,L)})$, the relational expert $f_{(r,R)}(d|\theta_{(r,R)})$, the numerical expert $f_{(r,N)}(d|\theta_{(r,N)})$ and the visual expert $f_{(r,I)}(d|\theta_{(r,I)})$. In addition, the variable c indexes all possible triples. We can combine the contributions above to produce the joint probability for a triple as follows:

$$p(d|\theta_1, \dots, \theta_n) = \frac{\prod_{F \in \{R,L,N,I\}} f_{r,F}(d|\theta_{(r,F)})}{\sum_c \prod_{F \in \{R,L,N,I\}} f_{r,F}(c|\theta_{(r,F)})}. \quad (5)$$

Some models incorporate a self-setting module in late fusion for improved performance. For example, the EVA [48] model uses an attention-weighted concatenation in modality fusion. Specifically, EVA applies the following equation to fuse features:

$$F_J = \bigoplus_{i=1}^n \left[\frac{e^{w_i}}{\sum_{j=1}^n e^{w_j}} \cdot F_i \right]. \quad (6)$$

In Equation (6), n is the number of modalities while w_i is the attention weight associated with the i th modality; F_i is the feature matrix corresponding to the i th modality (normalized in the L_2 norm). This weighted stitching approach enables EVA to learn the appropriate weights for each modality, effectively highlighting the most informative features in high-dimensional spaces.

The Knowledge perceived multi-modal pre-training Method in **E-commerce model (K3M)** [121] also uses a self-setting module to accomplish the fusion challenge. The K3M system considers three modalities, that is text, image, and knowledge (which describes relations and tail entities). The K3M system applies a co-attention Transformer [56] to capture interactions between the text and the imaging modality coupled with a fusion algorithm; to this end, K3M supports three types of fusion algorithms that is the mean of input feature vector and the soft-sampling and hard-sampling strategies outlined in [11]. Finally, the K3M system applies an improved GAT mechanism (similar to that proposed in [63]) to fuse the knowledge modality with text and image modalities.

Hybrid Fusion. Hybrid fusion is a novel approach that combines both early and late fusion, creating a more balanced and effective solution that addresses the shortcomings of each. To the best of our knowledge, the only system implementing hybrid fusion is DFMKE [118]. Early fusion is used in DFMKE through simple concatenation, which maps heterogeneous features to a shared space. However, early fusion alone can only ensure data consistency between different modalities and cannot guarantee the complementarity of data features. To address this issue, DFMKE utilizes a late fusion module called the LMF, which is efficient to leverage knowledge from multiple modalities [52]. If we denote as H_e and H_l the embeddings obtained by applying early and late fusion, respectively, DFMKE tries to maximize the agreement between H_e and H_l in a unified embedding

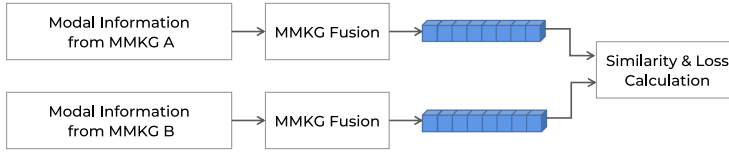


Fig. 6. The workflow of the alignment challenge.

space; to do so, it introduces the following loss function:

$$L = \sum_{i=1}^N \left(\|H_e^i - H_i^i Z^{(i)}\|_F^2 + \|I - Z^{(i)\top} Z^{(i)}\|_F^2 \right). \quad (7)$$

Here N is the number of entities while $Z^{(i)}$ is a matrix that maps the i th entity onto a shared space.

5.1.4 Alignment. Alignment is a crucial task in many domains, as it involves matching multiple entities or pieces of information that share the same semantic meaning. In MMKGs, the alignment task specializes as follows aligning entities from different MMKGs, aligning entities within the same MMKG, or aligning subcomponent modalities to a target entity. Since the same entity may be represented differently across different modalities or MMKGs, the alignment of entities is necessary for better knowledge discovery. For instance, aligning entities across multiple medical knowledge graphs can help researchers discover new relationships between diseases, treatments, and symptoms.

The workflow of MMKG alignment models generally follows a standard pattern, illustrated in Figure 6. Firstly, the model identifies the entities or subcomponents to be aligned. This can involve identifying entities with the same name or identifier across different MMKGs or selecting subcomponents (such as images or text) associated with a target entity. Next, the model generates embeddings or feature representations of the selected entities or subcomponents. These embeddings are high-dimensional vectors that capture the meaning of the entities or subcomponents. Entity/subcomponent meanings can be learned using neural networks or other machine learning techniques and can incorporate information from multiple modalities.

Once embeddings have been generated, the model computes a similarity score between the entities or subcomponents. The model may use various similarity metrics, such as the *cosine similarity* or the *Euclidean distance* to compute the similarity score. After computing the similarity score, the model optimizes a suitable loss function for training purposes. In most situations, the number of pre-aligned entity pairs is limited, making it challenging to achieve an ideal alignment performance. Thus, models often incorporate iterative learning modules for small-sample learning. These modules enable the model to learn from a small number of aligned pairs and gradually improve its performance over time. In the following, we list alignment models in Table 5 and describe how address the alignment task.

The MCTKG model [101] employed an entity alignment algorithm to fuse information from multiple data sources with the purpose of handling entities with missing attributes. We recall that the MCTKG model has been designed to operate in the tourism domain; for attraction entities, MCTKG aligned entities obtained from various online platforms with those obtained from the Beijing tourism website and combined their attribute values. The MCTKG model used an n -gram distance algorithm to perform the entity alignment, with each iteration aligning entity pairs with the minimum distance. However, this approach is difficult to train and not widely used in the field.

Due to the fact that entities to match are defined in different knowledge graphs, it is more convenient for us to verify if their representation vectors are closer in the same space. According to this line of reasoning, the MMEA model [12], after representing and fusing the relevant

Table 5. A Summary of Alignment Approaches in MMKG

Model	Similarity Measure	Loss Function	Iterative Learning	Optimizer
MCTKG [101]	N-Gram	–	√	–
MMEA [12]	Euclidean Distance	NCA	–	SGD
EVA [48]	Cosine Similarity	NCA	√	SGD
DFMKE [118]	Cosine Similarity	NCA	√	SGD
CLIP-Event [41]	Cosine Similarity	Kullback-Leibler Divergence	√	Adam

All methods under investigation handle structural, linguistic, and visual information.

information from multiple multi-modal knowledge graphs, creates a loss function based on the distances between aligned entities, as shown in Equation (8). Here E_1 and E_2 are entities from different multi-modal knowledge graphs. By minimizing the distances between aligned entity pairs, MMEA feeds back to the model training process, enabling the model to compute other potentially aligned entity pairs.

$$L_{ac}(E_1, E_2) = ||E_1 - E_2||_2^2. \quad (8)$$

EVA [48] incorporates an alignment learning component for aligning multi-modal embeddings. The model begins by computing the cosine similarity of all entity pairs and subsequently, it aligns them by minimizing an NCA loss [49] (see Equation (9)).

$$L = \frac{1}{N} \sum_{i=1}^N \left(\frac{1}{\alpha} \log \left(1 + \sum_{m \neq i} e^{\alpha S_{mi}} \right) + \frac{1}{\alpha} \log \left(1 + \sum_{n \neq i} e^{\alpha S_{in}} \right) - \log(1 + \beta S_{ii}) \right). \quad (9)$$

Observe that EVA leverages both local and global statistics to determine the importance of samples and penalizes hard negatives using a soft weighting scheme, aimed at addressing the so-called *hubness problem* in an embedding space [49]. EVA applies NCA loss separately for each modality and also to the merged multi-modal representation, as specified in Equation (10):

$$L_{joint} = \sum_i^n L_i + L_{Multi-modal}. \quad (10)$$

Additionally, the model employs an iterative learning approach to improving learning with minimal training pivots. Furthermore, the model introduces a probation technique that adds each pair of cross-graph entities to a candidate list, enhancing the stability of the process.

DFMKE [118] utilizes cosine similarity to measure the similarity between two entities. It aims at learning a function that maps the input entities into a low-dimensional space, where similar entities are mapped close to each other while dissimilar ones are far apart. To achieve this goal, the model uses the NCA loss function (see Equation (9)), which encourages the model to minimize the distances between similar entity pairs while maximizing the distances between dissimilar ones. DFMKE uses the **Stochastic Gradient Descent (SGD)** optimization algorithm. With a small batch size and a proper learning rate, DMFKE quickly converges to the optimal solution.

CLIP-Event [41] performs alignment at three levels: image-level, entity-level, and event-level. The model calculates cosine similarity between the text and image embeddings, as well as between the text entity and image object, at both the image-level and entity-level. It considers both mention similarity and type similarity. For the event-level alignment, the model applies optimal transport to find the shortest distance between the text event graph and image event graph. To train the alignment component, CLIP-Event uses Kullback–Leibler Divergence as the loss function and Adam as the optimizer function.

Table 6. Comparison of MMKG Models on Translation

Translation to Graph				
Model	Text Information			Visual Information
	Entity	Relation	Event	
GAIA [42]	LSTM-CRF attentive model	CNN-based model	Bi-LSTM CRF CNN-based model	Faster R-CNN
DCC [37]	SpaCy ¹³	Bi-LSTM	-	Diag2Graph
CLIP-Event [41]	LSTM-CRF attentive model	CNN-based model	Bi-LSTM CRF CNN-based model Dependency Parsing Tree	Faster R-CNN
MuMuQA [74]	OSCAR ¹⁴			OSCAR Faster R-CNN
Translation to other modalities				
Model	Translation Type	Method		
		Numerical	Text	Images
MKBE [71]	Generative	FFN	Adversarially Regularized Auto-Encoder	BE-GAN pix2pix-GAN

A horizontal bar means the model has got no lack of that modal information.

5.1.5 Translation. Translation is the process of converting information from one modality to another, making it more accessible across different modalities.

One important aspect of MMKG translation is the translation to a knowledge graph, also known as Local-MMKG. This involves extracting events or relationships from visual data, such as images or videos, and representing them in a structured format that can be easily integrated into a larger knowledge graph.

Another important aspect of MMKG translation is the translation to other modal information. This process involves generating or retrieving missing modal information according to the original entity embedding. As Li et al. [39] mentioned, there are two main types of translation to other modal information: *generative translation* and *example-based translation*. Generative translation creates new modal information from existing data, such as generating an image of an object from a text description. This can be useful if visual data is missing or incomplete, but textual data is available.

Example-based translation involves retrieving missing modal information from external sources, thereby establishing correspondences between different modalities. For instance, a retrieval conversion process may utilize an image of a person to retrieve additional information about them, such as their name or occupation, from a database or search engine. This approach can prove useful if visual data is available, but other modalities are either missing or incomplete.

Roughly speaking, translation provides additional context and information that can be used to enhance comprehension of the relationships between entities. By extracting events and relationships from different modalities, in fact, MMKG translation helps to build a more complete picture of the world and makes it easier to find and analyze information. In what follows we list models on translation in Table 6 describe in detail two types of translations, that is *translation to graphs* and *translation to other modalities*.

Translation to Graph. Translation to a graph involves extracting graph structures from pictures or texts, including relationship graphs or event graphs. Relationship graphs can represent the relationships between entities, while event graphs can represent the temporal and causal relationships between events. Translation to graph structure requires first extracting relevant entities

¹³<https://spacy.io>

¹⁴<https://github.com/microsoft/Oscar>

and relationships from text, extracting objects from the image structure, and then resolving co-references among these entities. Then, through the extracted relationships, they are connected to form a graph. By supplementing the modality information of MMKG with graph structures, translation to graph can help to obtain a more accurate and comprehensive knowledge representation.

GAIA [42] is a comprehensive open-source multimedia knowledge extraction system that performs information extraction from text. The model splits text information extraction into coarse-grained and fine-grained approaches. For entities, the model uses LSTM-CRF [45] for coarse-grained extraction and attentive models for fine-grained extraction. For extracting relations between entities, the model uses a CNN-based model. When it comes to event extraction, GAIA uses a combination of Bi-LSTM-CRF and CNN-based models for coarse-grained extraction, and verb/context/argument-based rules for fine-grained extraction. For visual information, the model uses Faster R-CNN [77] to detect objects. Once the extraction is complete, the model performs co-reference and entity linking by combining the entities and relations with two background knowledge bases, namely Freebase and GeoName. This process helps in creating a graph for a more comprehensive understanding of the extracted information.

Deep Code Curator. (DCC) [37] is a unified multi-modal knowledge graph constructed by extracting information from various scientific publications in the field of deep learning. The DCC model consists of three modules, namely Text2Graph, Image2Graph, and Code2Graph, each responsible for constructing different parts of the knowledge graph [110]. In Text2Graph, the model uses SpaCy to extract named entities and a Bi-LSTM with attention to extracting relations between entities. In Image2Graph, the model employs Diag2Graph to convert deep learning architecture diagrams into knowledge graphs. By integrating information from these modalities, DCC builds a comprehensive and interconnected knowledge graph. Finally, the model performs local and global graph alignment across the three modalities to ensure that the different parts of the graph are consistent with each other.

CLIP-Event [41] is a self-supervised contrastive learning framework that serves as a visual language model. As for text processing, CLIP-Event uses dependency parsing trees to extract primary events from captions and the aforementioned GAIA's extraction module to extract a wide range of newsworthy events. For visual information, the model uses Faster R-CNN to detect objects in Open Images. The model aligns the modalities of information to create MMKG, which is used to complete downstream tasks.

From the models mentioned above, we can see that there are several powerful techniques that can be applied to graph translation. For text extraction, CRF is particularly effective. More specifically, CRF takes into account the dependencies between neighboring words in a sentence and uses them to make more accurate predictions. CNN are an appealing option to detect relationships. In addition, sequence-based networks are crucial for accurately extracting information from text. Sequence-based networks, in fact, can model long-term dependencies between words in a sentence. The **Fast R-CNN (FRCNN)** algorithm is a popular choice for visual-based extraction. Specifically, it has been shown to be effective in a wide range of visual-based text extraction tasks, such as license plate recognition, OCR, and text detection in natural scenes.

Translation to other Modalities. Translation to other modalities involves transforming information from one modality to another while preserving its original meaning. Specifically, we first decode the entity embedding, which is a condensed representation of the information in the original modality, and we map it onto the target modality using techniques such as neural networks and probabilistic models.

After decoding the data, translation to other modalities can be divided into two categories: generative translation and example-based translation. The generative translation is focused on producing novel modal information by decoding existing data embeddings to generate image descriptions

or video captions. On the other hand, example-based translation aims at retrieving absent modal information from external sources. This method involves using examples from the target modality to fill in missing information in the decoded entity embedding, although it's uncommon in MMKGs. Despite being similar to multi-modal learning translation, translating to other modalities still has a significant role in MMKG. In the local MMKG model, this technology assists in gathering diverse modal information for constructing an all-inclusive MMKG. Similar to translating into a graph, translating to other modalities also enriches the modality information of MMKG with various modalities, resulting in a more extensive MMKG.

To the best of our knowledge, the only existing model that includes translation to other modalities is MKBE. MKBE is a powerful model that can decode entity embeddings into various modalities of information, including numerical and categorical data, text, and images. To predict numerical and categorical data, the model uses a simple feed-forward network with RMSE and cross-entropy loss functions. When it comes to text, MKBE leverages an **Adversarially Regularized Auto-Encoder (ARAE)** to generate relevant sentences. For generating high-quality images, the model uses a combination of BE-GAN structure and pix2pix-GAN. Overall, MKBE is a versatile and efficient tool for processing and generating various types of data.

5.2 Pre-training in MMKGs

The primary aim of MMKG construction or its downstream tasks is to integrate additional modal information and facilitate reasoning. In our investigation of existing MMKG models, we observed that pre-training models effectively address two significant challenges: effectively managing the rich information contained in MMKG and supporting a wide range of downstream tasks. Specifically, pre-training models are utilized for information extraction [106], information representation [93, 106, 121], and downstream task training [74, 121]. The pre-training process involves training machine learning models on extensive datasets, often unsupervised, to acquire general-purpose representations that can be fine-tuned for specific tasks. After undergoing pre-training on extensive datasets, models have demonstrated effectiveness in specific tasks. When faced with the aforementioned challenges, pre-training models can be fine-tuned (or sometimes even directly applied) to achieve desired outcomes. This approach allows us to save time and effort when dealing with the vast amount of data in MMKG. Pre-training models offer not only efficiency but also versatility. Specifically, pre-training models such as BERT [18], GPT [72], RoBERTa [51], and REALM [26] have demonstrated their effectiveness in various NLP-related tasks. These models leverage large-scale natural language corpora, often extracted from multiple web sources, and have significantly advanced the state-of-the-art in many NLP tasks. Additionally, pre-training models can facilitate knowledge graph completion [54, 55, 109], as well as generate text solely based on the information contained within the knowledge graph.

We believe that pre-training models have the capability to efficiently handle MMKG-related problems, and their significance in MMKG is highly foreseeable. Therefore, we dedicate a separate chapter to discussing pre-training in MMKG. In the following, we provide a detailed description of its usage and the benefits it brings to each component of the MMKG life cycle.

5.2.1 Pre-training in Feature Embedding. Feature embedding is a commonly used technique for representing data in downstream tasks. This section focuses on the pre-training in feature embedding in MMKG, which can be divided into *text* and *image embeddings*. Since no pre-trained model has been used for graph structure embedding in MMKGs so far and most MMKG models rely on a scoring function to accomplish graph structure embedding, we will not elaborate on the pre-training in graph structure embedding.

Pre-training for text embedding has become increasingly popular in recent years. For instance, K3M [121] uses WordPieces based on BERT to convert text into token sequences, which are then

Table 7. Application of Pre-trained Models in the MMKG

Pre-traning in Feature Embedding			Pre-traning in Downstream Tasks		
Model	Text	Visual	Model	Tasks	Method
K3M [121]	WordPieces	VGG19	K3M [121]	MOM/MLM/LPM	<i>itself</i>
Alime MKG [106]	StructBERT	ViT	MuMuQA [74]	Question Generation	BART
Richpedia [93]	–	VGG16			
Pre-training in Assistance					
Assistance of Construction			Assistance of challenges		
Model	Goal	Method	Model	Challenges	Method
CLIP-Event [41]	Auxiliary Text Generation	GPT-3	K3M [121]	Fusion	co-attention Transformer
Alime MKG [106]	Text Knowledge Extraction	BERT	Alime MKG [106]	Fusion	two-stream Transformer
			GAIA [42] CLIP-Event [41] MuMuQA [74]	Translation	Faster R-CNN

put into the encoder based on Transformer. Similarly, Alime [106] uses structBERT to encode textual information. Pre-trained models for text embedding can learn general language representations from a large amount of data; in this way, pre-trained models are useful for a broad range of natural language processing tasks, often outperforming models trained from scratch. Moreover, pre-trained models can handle multiple languages and transfer learning from one language to another, thereby accelerating progress in cross-lingual natural language processing.

Visual information embedding, unlike text, requires a pre-training model with a large amount of example data to support high-quality image embedding. Pre-trained models can learn universal visual features from massive image datasets, avoiding the tedious, expensive, and time-consuming process of training from scratch. Popular pre-training models are VGG, ResNet, and ViT, and they are frequently used in MMKG with ideal performance. The VGG model has a simple network structure that is easy to implement and understand, while the ResNet model uses residual blocks to build the network, making it easier to train. Some visual pre-training methods used in MMKG construction include Alime MKG [106] using ViT based on Transformer and Richpedia using VGG16 to embed visual information.

5.2.2 Pre-training in Assistance. Pre-training is often used to construct an MMKG as well as to deal with the four challenges we introduced before. Although some pre-training techniques may not directly impact the model's overall performance, they can help a model complete its tasks. We categorize pre-training into *assistance of construction* and *assistance of challenges*.

Assistance of Construction. Pre-trained models provide a concrete benefit to constructing MMKGs, as we will show in this section. Specifically, models such as CLIP-Event [41] and Alime MKG [106] are relevant examples of how pre-training can be used to generate or extract useful information.

CLIP-Event [41] utilizes GPT-3 to generate additional text based on five manually written event description examples. The pre-trained model is fine-tuned to produce similar descriptions for other events, aiding in the creation of knowledge graphs. CLIP-Event utilizes GPT-3's natural language processing to generate precise and contextually relevant descriptions, enhancing MMKG accuracy and completeness.

Alime MKG [106] uses a binary BERT classifier to classify extracted phrases and complete text knowledge extraction. By using this approach, Alime is able to identify key phrases and concepts from a given text and convert them into knowledge graph nodes and edges. This not only makes the extraction process more efficient but also enables the creation of more comprehensive knowledge graphs.

Assistance of challenges. Pre-training is also commonly used to assist with the four fundamental challenges we previously introduced, that is representation, fusion, translation, and downstream tasks.

In the representation challenge, the MARS model [115] pre-trains on the MARKG dataset using IKRL [104], TransAE, and RSME as reference multi-modal knowledge graph embedding methods to obtain entity embeddings.

Pre-trained models are mostly used to address the fusion challenge. The key issue in fusion is *modal interaction*, that is, how to bring together information from multiple modalities. K3M uses co-attention Transformer to complete the fusion process, while GAKA [16] uses BERT to fuse each paper entity with annotated article abstracts. Alime MKG [106] uses a two-stream Transformer structure from LightDOT to fuse texts and images. Some models use pre-training not to unify text, images, and graph structures all at once, but they have to complete pairwise fusion first before moving on to multi-modal fusion. This assistance is actually the migration of some mature fusion techniques from multi-modal learning or knowledge graph learning to the MMKG model, improving the overall fusion effect.

Pre-trained models have also been applied to meet the translation challenge. As shown in Table 7, most models (including GAIA [42], CLIP-Event [41], and MuMuQA [1]) use Faster R-CNN [76, 77] for object detection purposes. The ability to detect objects accurately helps improve the accuracy of subsequent co-reference resolution tasks and enhances the overall quality of MMKG.

5.2.3 Pre-training in Downstream Tasks. Pre-trained models are widely utilized for completing downstream tasks, with MuMuQA [74] using fine-tuned BART [38] as an encoder-decoder to train models for natural language question generation. Originally developed for text generation tasks, BART is a pre-trained model that MuMuQA fine-tunes for question generation by providing it with question-answer pairs as training data. Similarly, the K3M model [121], introduced earlier, is a pre-trained multi-modal model that leverages knowledge from different modalities to correct noise and supplement missing information, relying on fine-tuning to complete several downstream tasks such as multi-modal sentiment analysis and multi-modal question answering. Pre-trained with three tasks - **Masked Object Modeling (MOM)**, **Masked Language Modeling (MLM)**, and **Link Prediction Modeling (LPM)** - the K3M model effectively combines information from different modalities and captures rich semantic information from multiple sources, including images, text, and knowledge graphs, making it possible to fine-tune for specific downstream tasks. With state-of-the-art results across a range of multi-modal tasks, the K3M model's success demonstrates the power of pre-training in multi-modal learning.

6 DOWNSTREAM TASKS, APPLICATION FIELD AND EVALUATION CRITERIA

Downstream tasks refer to the actual tasks that can be performed with the help of an MMKG. The list of potential downstream tasks is quite extensive and includes tasks such as classification, recommendation, prediction, and generation. We categorize downstream tasks into *generative* and *non-generative* tasks. We provide a detailed description of each category along with the evaluation metrics used for the experimental assessment of existing techniques.

6.1 Generative Tasks

In generative tasks, a model is required to generate new output based on given input information. Relevant examples of generative tasks are text summarization, image generation, and question answering. MMKGs have been utilized to support **Question Answering (QA)**, where a system can comprehend the question's meaning and locate relevant information to provide a correct and concise answer. Compared to other models, the MMKG model combines image information with event

relationships, providing more reference information and enhancing answer accuracy. For instance, MUMUQA [74] proposes a multi-hop question decomposition model that performs multi-hop reasoning on the question, title, and body. This approach consistently outperforms using multi-hop reasoning question answering or end-to-end multi-modal question answering systems separately, highlighting the advantages of MMKGs. In QA tasks, the generative model is typically evaluated using metrics such as BLEU and METEOR, which are discussed below.

BLEU. The BLEU (Bilingual Evaluation Understudy) metric [68] evaluates the quality of machine translation by measuring the similarity between the generated translations and reference answers. BLEU scores range from 0 to 1, with higher scores indicating better translation quality.

METEOR. METEOR (Metric for Evaluation of Translation with Explicit ORDERing) [5] completes the evaluation by comparing the similarity between machine translation and reference answers in terms of words, phrases, forms, and order, making it more comprehensive and accurate compared to BLEU. METEOR scores range from 0 to 1, with higher scores indicating a better translation quality.

6.2 Non-generative Tasks

Non-generative tasks are those tasks where the output is not generated from scratch but derives from existing data. Examples of non-generative tasks include classification, recommendation, and prediction.

Multi-modal information jointly with the graph structure of the MMKG acts as a kind of *contextual information* and, thus, play a crucial role in enhancing the accuracy. As an example for the recommendation task, the MKGAT system [89] introduces a recommendation module that takes advantage of both the knowledge graph embedding of entities and the collaborative knowledge graph as inputs, enhancing the representation of both users and items. Then, MKGAT applies formal recommendations to compute matching scores between users and items. As a further example, MKRLN [91] proposes a framework for explaining recommendations over the MKG, effectively reasoning the logical paths of visuals and knowledge for each user leading to the recommended items using reinforcement learning.

A further important example of a non-generative task is the detection of (suspicious) events or trends from observed data as well as the related task of predicting possible outcomes. A relevant example is given by *fake news detection*; here, multi-modal information can improve the model's performance as a semantic information supplement. KMGCN employs a multi-modal graph convolutional network to aggregate the nodes of each graph and obtain representation vectors for each post. It extracts the semantic-level features of each post and feeds the multi-modal representation of each post into a binary classifier, and then determines whether the news is fake by calculating the predicted probability.

Analogy reasoning refers to the process of comparing two or more different objects with the purpose of, finding their similarities and relationships; obtained similarities and relationships are exploited to solve new problems or understand new objects. The MART model [115] is a cross-analogy interactive encoding process based on Transformer that adaptively interacts between analogy examples and question-answer pairs.

Non-generative tasks are commonly evaluated using metrics as Hit Ratio, RMSE, F1 score, mAP, and NDCG@n. In the following, we briefly describe each of these metrics.

Hits@K. Hits@k is a commonly used evaluation metric in information retrieval and recommendation tasks, which evaluates the number of hits or relevant items retrieved by the model in the top-k recommendations. Hits@k takes into account the order of recommendations and rewards the model for presenting relevant items at the top of the list.

RMSE. The **Root Mean Square Error (RMSE)** is the expected value of the squared difference between the parameter estimation value and the true parameter value, with smaller value indicating better performance.

MR/MRR. MR and MRR are performance metrics commonly used in information retrieval systems. MR is calculated by taking the average reciprocal of the rank of the first relevant document for each query, whereas MRR calculates the average reciprocal of the rank of all relevant documents for each query. MRR is considered to be a better metric since it considers the impact of relevant documents ranked lower in the list.

F1 score. The F1 score is a metric used to evaluate the performance of binary classification models. It considers both the accuracy of predictions and the ability to correctly identify positive samples, quantifying the overall effectiveness of the model by calculating the harmonic mean of precision and recall.

mAP. The **Mean Average Precision (mAP)** is a crucial metric in the field of target detection, which relies on both the F1 score and the **Intersection over Union (IoU)** intersection ratio. The mAP represents the average accuracy of the entire class and is typically calculated as the weighted average of the **Average Precision (AP)**.

NDCG@n. **Normalized Discounted Cumulative Gain at n (NDCG@n)** is a widely adopted measure used to assess the accuracy and quality of ranked lists in information retrieval and recommendation systems. This metric assigns scores between 0 and 1 to each item based on its position in the list, with higher scores indicating better performance.

6.3 Quality Verification Task

Multi-modal quality verification is relatively new and rare. It aims at providing a baseline model to evaluate the quality of the multi-modal dataset in accomplishing downstream tasks.

6.3.1 Multi-modal Quality. GEM. GEM [87] is a large-scale multilingual multi-modal benchmark consisting of GEM-I for image language tasks and GEM-V for video language tasks. As the largest visual language dataset, GEM covers both image and video language tasks and it can be used to evaluate the generalization ability of visual language pre-trained models on a diverse set of multi-modal tasks. GEM provides two multilingual and multi-modal pre-trained models, M3P [66] and m-UniVL [58], as baselines for image-language and video-language tasks, respectively.

6.3.2 Knowledge Quality. CKRL. CKRL [102] is an advanced framework for knowledge representation learning that integrates the notion of confidence into conventional translation-based methodologies. This model leverages the intrinsic structural details of KGs to introduce three distinct types of triple confidences, which take into account both local and global structural characteristics. By incorporating these enhanced triple confidences, CKRL enhances flexibility and universality in capturing the confidence levels of knowledge triples.

KGTtm. KGTtm [35] is a computational framework designed to assess the trustworthiness of knowledge graph triples, quantifying the semantic accuracy and veracity of the information represented within the knowledge graph. This model adopts a cross-neural network architecture, wherein the upper layer incorporates a trustworthiness evaluation unit consisting of a resource rank algorithm, translation-based energy function, and reachable paths inference algorithm. The lower layer is the fusion unit, which generates the final trustworthiness value for each triple.

7 EXISTING PROBLEMS AND FUTURE DIRECTION OF MMKG

In this section, we first list the existing problems of MMKG models. Then, we discuss the portability of multi-modal learning techniques, which aims at providing a reference for the improvement of

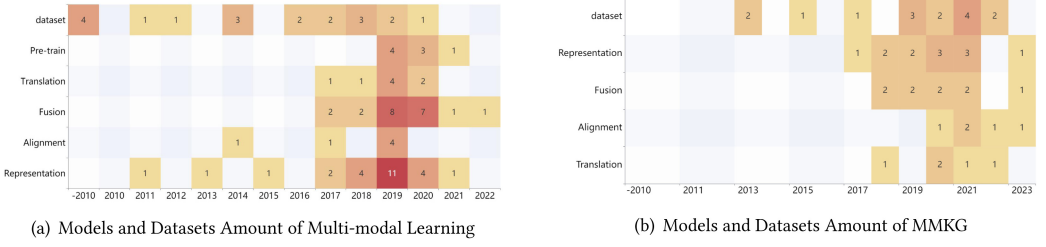


Fig. 7. Heatmaps of multi-modal learning and MMKG.

MMKG. Ultimately, we point out the future direction of MMKG, hoping to be helpful to related researchers.

7.1 Existing Problems in MMKG

The construction and management of MMKGs is a relatively new and rapidly evolving research area. Despite significant advances in recent years, MMKG is still not considered as a mature technology if compared to other related fields such as multi-modal learning and knowledge graph learning. Hence, the prevailing presence of unresolved issues and challenges necessitates diligent efforts to be undertaken in order to fully actualize the potential of MMKG.

Small Amount of Existing MMKGs. The limited number of existing MMKG models and datasets indicates that MMKG is still in its early stages. As shown in Figure 7, there are significantly fewer MMKG models than multi-modal learning models. Currently, there are only 18 MMKG models (some of which are involved in different challenges simultaneously) and 15 publicly available datasets developed and made available for research purposes.

Explainability and Transparency of the Feature Fusion Process. Unlike traditional knowledge graphs, which are primarily text-based, MMKGs incorporate multimedia content. The availability of heterogeneous resources poses significant challenges in terms of representing and integrating these different types of data into a cohesive knowledge graph. Due to the complexity of the feature fusion process, the output of a system driven by an MMKG could be difficult to understand by the end user. Though significant research efforts aimed at uniformly representing heterogeneous data instances, it is crucial to develop more effective algorithms and techniques for reasoning over multimedia data.

Challenging for Fulfilling Diverse Downstream Tasks. The ability to exist MMKG datasets to support diverse downstream tasks is significantly limited by two primary factors: the datasets themselves and the model architecture employed. With respect to the datasets, as illustrated in Table 2, half of the existing MMKG datasets are specialized for specific domains. Their limited scope makes them inadequate for more common tasks. Even the more general-purpose MMKG datasets, with the exception of Richpedia, are often quite small in size. The small size of available datasets drastically restricts their ability to support diverse downstream tasks, particularly those tasks that require large amounts of training data. As for the model architecture, most MMKG models are designed and trained for a single downstream task, further restricting their applicability to other types of downstream tasks.

Suboptimal Downstream Task Performance. The goal of MMKG is to enhance the performance of a knowledge graph by incorporating rich multi-modal information into its structure. However, tasks such as entity alignment in MMKGs are intrinsically hard, and, for instance, the accuracy rate of MMKG alignment is around 50%, which is lower than that of multi-modal learning. Consequently, the accuracy of MMKG downstream tasks is generally lower than expected. We

believe that this situation may arise due to semantic differences and expressive variations between different modal data, namely, modality heterogeneity or semantic gaps.

Higher Complexity of Calculation. Compared to pure multi-modal or knowledge graph computations, MMKG results in a significant increase in computational complexity and workload. This heightened complexity arises primarily from the intricate relationships between integrating multi-modal data and knowledge graph information, making the computation process more expensive.

7.2 Future Direction

As a relatively new field, MMKG has a vast potential for future development in diverse directions. We are confident that MMKGs will become a mainstream direction in the future. How to develop MMKG is a worthy topic of discussion. We believe that the development of MMKG involves: (i) *adding datasets*; (ii) *leveraging mature technologies*; (iii) *pre-training*; (iv) *incorporating logical reasoning*; (v) *exploring new attributes and modalities*.

Adding datasets. Addressing existing problems is crucial. As discussed in Section 7.1, these problems are mainly caused by the limited number of data sets and the inflexibility of model architectures. One approach to solve this issue is to create additional MMKG data sets. Researchers can utilize more flexible tools to gather and construct MMKG data sets. For instance, OpenIE [21], an open extraction tool, can aid in collecting entities and relationships from a data set. The expansion of the number and size of data sets combined with innovative neural architectures is likely to enhance the performance of downstream tasks.

Leveraging Mature Technologies. Due to the long history and extensive research in the fields of multi-modal learning and knowledge graph learning, there are many existing models, and the technology is relatively mature and performs well. We can transfer mature technologies to the MMKG. For example, the pre-training model ViT can efficiently integrate text and images in MMKGs. Pre-training can help improve the accuracy, robustness, and efficiency of MMKG models, making them more effective for various downstream tasks.

Pre-training. We believed that pre-training is a very important technique for MMKG. By using as much training data as possible and extracting common features, pre-training aims at reducing the learning burden of specific tasks and achieve better performance when training on less or smaller datasets. At a macro-level, we believe that MMKG data is usually large in scale, so it is necessary to reduce workload through pre-training. Models like K3M complete all tasks using only pre-trained models. This approach not only saves a lot of time and effort, but also makes the model structure more flexible, and it makes the model able to complete multiple downstream tasks through fine-tuning. At a micro-level, we believe that parts of some models can also be processed using pre-trained models. For information extraction, we can draw on the UIE model [57]. For text extraction tools, we can use EIMo or GPT models. And for embedding the knowledge graph structure, we can refer to the KEPLER model [94], which provides concrete benefits in various tasks.

Incorporating Logical Reasoning. We believe that multi-modal knowledge graphs will become mainstream in the future. Incorporating logical reasoning can make MMKGs more consistent with human thinking. We can add to an MMKG various logical reasoning methods, such as Semantic Web Reasoning, Neural Network Reasoning, and Graph Neural Network Reasoning. Adding logical reasoning to MMKG can not only deduce new knowledge and discover hidden relationships for better understanding and discovery of knowledge but also increase its scalability, making the MMKG more powerful and applicable.

Exploring New Attributes and Modalities. We believe that adding more attributes or modalities will enrich the information in MMKG, creating a large dataset with multiple downstream tasks. For instance, the model can reference the cascading global topology structure of social

network information and incorporate temporal information, allowing models with event-relatedness to capture internal connections. Such additions transform MMKG from a fixed knowledge graph into a dynamic event graph with temporal and causal aspects. Moreover, additional modalities enhance the knowledge representation capability of KG, enabling it to better describe the real world and aiding in more accurate reasoning and prediction.

8 CONCLUSIONS

This survey provides a comprehensive review of the literature in the emerging field of MMKG. While multi-modal machine learning is an established research area, knowledge graphs are a dominant technology that has enabled the development of sophisticated artificial intelligence applications. In MMKGs, we have the ability to represent knowledge in a structured manner and leverage complementary signals such as text and images to provide deep and accurate representations of knowledge. As a result, MMKGs are effective in handling complex queries and offer opportunities beyond those provided by traditional knowledge graphs. Additionally, MMKGs can use various signals, such as images available on the internet, to determine whether two concepts extracted from texts in different languages and encoded at different levels of accuracy refer to the same real-world entity.

Despite their vast potentialities, there is no generally accepted definition of MMKGs and, to fill this gap, we provided in this article a theoretical explanation of the concept of MMKGs and we also illustrated the main steps in the MMKG construction process, specifying how some real-world systems implement each of these steps. We then addressed the problem of how to improve an MMKG through reasoning. We identified four fundamental challenges (representation, alignment, translation, and fusion) that are relevant to enrich an MMKG as well as to make it more accurate. The identification of four fundamental challenges immediately provides a *flexible classification scheme* of existing approaches: indeed, we have analyzed how real systems deal with one or more of these challenges, and as a result, we are able both to present and compare approaches in the literature in a unified way as well as to easily incorporate future approaches. During our discussion, we placed a particular emphasis on pre-training models, which have become increasingly popular for representing both textual and visual content. We believe that pre-training models can serve as a significant catalyst for advancing the state of the art in MMKG research. Furthermore, we defined the metrics commonly used in the literature to assess the performance of existing MMKGs in various downstream tasks. Finally, we delved into open problems and proposed potential solutions to these issues. Overall, we believe that our article provides valuable inspiration and guidance for researchers in this field, and establishes a strong foundation for future studies in multi-modal learning.

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